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SYNTHESIS, CHARACTERIZATION AND EVALUATION OF ANTIBACTERIAL ACTIVITY OF AZADIRACHTA INDICA, ACALYPHA INDICA LOADED ZIF-8 NANOCOMPOSITES

PROJECT REPORT

Submitted to

PSGR KRISHNAMMAL COLLEGE FOR WOMEN

In partial fulfillment of the requirements for the award of the degree of

MASTER OF SCIENCE IN CHEMISTRY

Submitted by

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MARCH 2026

CERTIFICATE FROM GUIDE

This is to certify that **Ms. S. Karthika (24SMCH009)**, a II- year M.Sc. Chemistry student of the Department of Chemistry at PSGR Krishnammal College for Women, Coimbatore - 641004, has successfully completed her project work titled “**Synthesis, Characterization and Evaluation of Antibacterial Activity of Azadirachta indica & Acalypha indica Loaded ZIF-8 Nanocomposites**” under the guidance of Dr. G. R. Rajarajeswari, Associate Professor, Department of Chemistry, Anna University, Chennai - 600025, for a period of 3 months from December 2025 to March 2026.

**EXTERNAL GUIDE
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Dr. M. Dharmendira Kumar

CERTIFICATE

This is to certify that the project entitled “**Synthesis, Characterization and Evaluation of Antibacterial Activity of Azadirachta indica & Acalypha indica Loaded ZIF-8 Nanocomposites**” submitted to PSGR Krishnammal College for Women, Coimbatore, in partial fulfilment of the requirement for the award of the Degree of **Master of Science in Chemistry** is a record of original project work done by **Ms. S. Karthika (24SMCH009)** at Department of Chemistry, Anna University, Chennai – 600025 during the period 2025 - 2026, under the supervision and guidance of **Dr. G. Subashini**, Assistant Professor, Department of Chemistry, PSGR Krishnammal College for Women, Coimbatore and **Dr. G. R. Rajarajeswari**, Associate Professor, Department of Chemistry, Anna University, Chennai - 600025. This project work has not formed the basis for the award of any degree, fellowship, diploma or other similar title.

**Signature of the Supervisor
(Internal)**

**Signature of the Supervisor
(External)**

Signature of the HOD

Signature of the Principal

External Examiner

DECLARATION

I, **KARTHIKA S (24SMCH009)** hereby declare that the project entitled “**Synthesis, Characterization and Evaluation of Antibacterial Activity of Azadirachta indica & Acalypha indica Loaded ZIF-8 Nanocomposites**” submitted to PSGR Krishnammal College for Women, in partial fulfilment of the requirement for the award of the Degree of Master of Science in Chemistry is a record of the original project work done by me at PSGR Krishnammal College for Women, Coimbatore during the study period of DECEMBER 2025 - MARCH 2026 under the supervision and guidance of **Dr. G. Subashini**, Assistant Professor, Department of Chemistry, PSGR Krishnammal College for Women, Coimbatore and **Dr. G. R. Rajarajeswari**, Associate Professor, Department of Chemistry, Anna University, Chennai - 600025. This dissertation has not formed the basis for the award of any degree, fellowship, diploma or other similar title.

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ACKNOWLEDGEMENT

“I owe my sincere gratitude to Almighty, who has led me throughout my life,”

I have great pleasure in expressing my sincere thanks to the management, **Dr. N. Yesodha Devi**, Secretary and **Dr. P.B. Harathi**, Principal PSGR Krishnammal College for Women, Coimbatore, for giving me suggestions and permissions to do the project work at.

I am grateful to **Dr. D. Nalini**, Head and Associate Professor, Department of Chemistry, PSGR Krishnammal College for Women, Coimbatore, for her valuable guidance and encouragement throughout the period of my study.

I would specially thank **Dr. M. Dharmendira Kumar**, Professor and Head, Department of Chemistry, Anna University, Chennai for permitting me to do my project.

I have extremely immense pleasure in expressing my jubilant feeling and whole hearted gratitude to **Dr. G. R. Rajarajeswari**, Associate Professor Department of Chemistry, Anna University, Chennai for her constant encouragement and help rendered throughout the project work.

I record my special and sincere thanks to **Dr. G. Subashini**, Assistant Professor, Department of Chemistry, PSGR Krishnammal College for Women, Coimbatore, for her support throughout my project work.

I express my grateful thanks to entire teaching faculty, Department of Chemistry, PSGR Krishnammal College for Women.

I express my sincere thanks to **Ms. R. Aseela Banu** Research Scholar, Anna University, Chennai for their caring and enthusiastic guidance, encouragement and constructive idea for the successful completion of the project work.

I take this opportunity to express my sincere thanks to my family for their encouragement and friends who have helped throughout this project work.

KARTHIKA S

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ABSTRACT

Metal-organic frameworks (MOFs) are a class of porous materials that have gained significant attention due to their unique structural properties such as high surface area, tunable pore size, and good chemical stability. Among them, zeolitic imidazolate framework-8 (ZIF-8) is widely studied for various biomedical applications. In the present work, ZIF-8 was synthesized using zinc nitrate hexahydrate and 2-methylimidazole by a simple solution-based method. Medicinal plant extracts of *Azadirachta indica* and *Acalypha indica* were prepared using the Soxhlet extraction method. These extracts were then incorporated into the ZIF-8 framework to form plant extract-loaded nanocomposites, namely AZ@ZIF-8 and AC@ZIF-8. The formation of these materials was initially indicated by a change in colour during synthesis. The synthesized samples were characterized using FTIR, XRD, SEM, and TGA techniques to study their functional groups, crystal structure, surface morphology, and thermal stability. The results confirmed the successful formation of ZIF-8 and the incorporation of plant extracts within its structure. The antibacterial activity of the prepared materials was tested against *Escherichia coli* and *Staphylococcus aureus*. The plant extract-loaded ZIF-8 samples showed better antibacterial activity compared to pure ZIF-8 and individual plant extracts. This may be due to the combined effect of zinc ions and bioactive compounds present in the plant extracts, which help in inhibiting bacterial growth. Thus, the present study demonstrates that plant extract-loaded ZIF-8 can be considered as a promising material for antibacterial applications and may have potential use in biomedical fields.

CHAPTER – 1

INTRODUCTION

1.1 MOFs:

Coordination polymers are a class of compounds formed through the coordination of metal ions with inorganic or organic ligands, resulting in extended frameworks stabilized by coordination bonds. Owing to their tunable structures and multifunctional properties, these materials have been extensively investigated and have inspired the synthesis of a wide range of related compounds. Among them, metal–organic frameworks (MOFs) represent a distinctive subclass characterized by porous architectures constructed from the coordination of metal ions or metal clusters with organic ligands. The unique structural features of MOFs, particularly their high porosity and design flexibility, have attracted significant attention in recent years, positioning them as promising materials for diverse applications [1]. Metal–organic framework (MOF)-based materials, encompassing pristine MOFs, MOF composites, and MOF-derived structures, have emerged as a rapidly expanding class of functional materials due to their distinct advantages over conventional systems in energy conversion and storage applications. Pristine MOFs, often referred to as porous coordination polymers, are typically synthesized through the self-assembly of metal nodes-such as isolated ions or polynuclear clusters-with organic ligands via strong coordination bonds. This synthetic strategy imparts MOFs with highly ordered porous architectures, exceptional structural diversity, and tunable functionality. As a result, pristine MOFs exhibit remarkable properties, including high surface area, adjustable pore environments, and chemical versatility, which collectively underpin their superiority and widespread applicability in advanced energy-related technologies [27]. The synthesis of nanoscale metal–organic frameworks (MOFs) has been extensively explored owing to their potential applications in sensing and drug delivery. In recent years, the development of hybrid porous materials has accelerated, driven by the need to optimize structural features, chemical compositions, and functional properties. MOFs, also referred to as porous coordination polymers (PCPs), represent a unique class of composite inorganic–organic crystalline materials characterized by persistent porosity. Their nanoscale dimensions and exceptionally high surface area enhance bioactivity, improve circulation, enable surface modification, and contribute to superior chemical and colloidal stability. The versatility of MOFs arises from the ability to incorporate a wide range of chemically active components, allowing the design of nanostructures with tailored

characteristics. This adaptability has positioned MOFs as intriguing candidates for use as antibiotic supplements or alternatives. Structurally, MOFs are composed of metal clusters coordinated with organic linkers, and their crystalline architecture, combined with tunable linker length, governs porosity and surface area—key attributes that underpin their effectiveness in catalysis. Beyond catalysis, MOFs have demonstrated broad utility in domains such as gas storage, adsorption, sensing, and antibacterial activity. The synergistic contribution of their organic and inorganic components further enhances bacterial eradication, reinforcing their potential as advanced antibacterial materials [2].

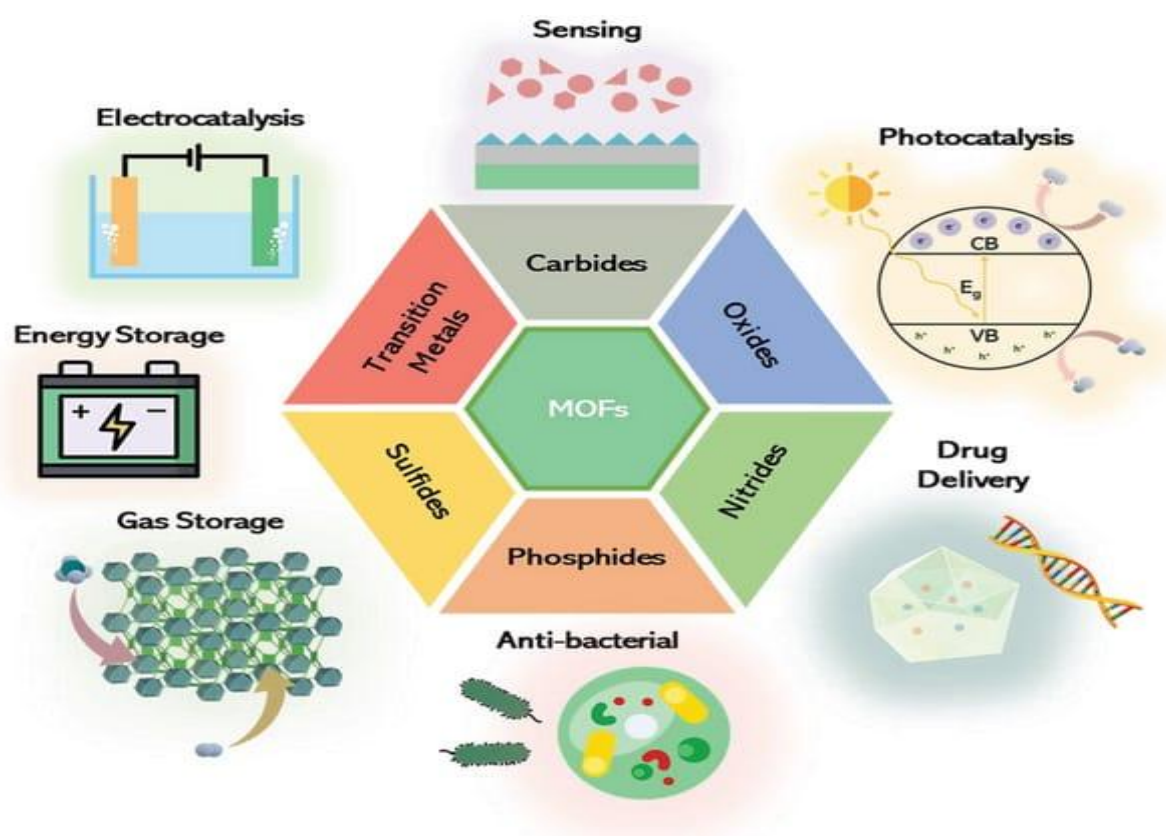


Figure 1.1 MOFs types and application [27]

Compared with pure inorganic molecular sieves and porous carbon materials, MOFs have the following advantages:

- (1) The highly crystalline state of MOFs is very conducive in determining their precise spatial structure with single crystal and polycrystalline diffraction methods.
- (2) MOFs have high porosity and a specific surface area.
- (3) MOFs can be composed of various different metal ions and organic bridging ligands, and they have structures that are easy to design.

(4) The σ single bond in the organic ligand structure gives MOFs a certain degree of flexibility, so that they have characteristic functions.

(5) The structure is easy to modify: MOFs frame and the structure of the pore surface can be adjusted by modifying the metal centre and organic ligands in the MOFs framework so as to give MOFs a variety of functions.

1.1.1 Types of MOFs:

Divalent metal ions, particularly those from the first transition series such as Mn^{2+} , Co^{2+} , Cu^{2+} , and Zn^{2+} , are frequently employed in the construction of MOFs and have proven effective as drug carriers. According to the principles of the hard–soft acid–base (HSAB) theory, these metals possess an appropriate balance of softness and hardness, enabling coordination with common donor atoms such as oxygen and nitrogen with moderate reversibility. This reversible coordination imparts MOFs with a distinct advantage in drug delivery, as drug release can be triggered under specific stimuli such as changes in pH or temperature, which weaken the coordination bonds and facilitate controlled release. In contrast, monovalent ions including Cu^+ , Ag^+ , and K^+ are also utilized in MOF synthesis, but their sensitivity to light, heat, and moisture often compromises the stability of the resulting frameworks, necessitating the use of carefully selected ligands to achieve acceptable structural robustness. Trivalent metal ions such as Cr^{3+} , Fe^{3+} , and Al^{3+} typically form coordination bonds with oxygen-containing ligands, particularly carboxylates, that exhibit stronger covalent character. While this enhances the chemical and thermal stability of the resulting MOFs, it also promotes side reactions during synthesis, such as the formation of hydroxyl- or oxygen-bridged polynuclear clusters, which can hinder framework assembly and growth. Consequently, the preparation of MOFs based on Cr^{3+} , Fe^{3+} , or Al^{3+} often requires highly acidic conditions and elevated reaction temperatures. Similarly, tetravalent metal ions such as Zr^{4+} form coordination bonds with oxygen donors that are strongly covalent in nature, demanding more severe reaction conditions for successful MOF synthesis. The stability of coordination bonds between metal ions and organic ligands, coupled with the tunability of ligand design, plays a crucial role in the synthesis of MOFs. Carboxylate and pyridine ligands are among the most widely employed linkers. Carboxylates exhibit strong binding affinity toward trivalent and tetravalent metal ions, and their negative charge effectively neutralizes the positive charge of metal centers or clusters, thereby enhancing both porosity and

framework stability. However, the diverse coordination modes of carboxylates often make their structural outcomes difficult to predict and control. In contrast, pyridine ligands coordinate in a simpler manner, but their lack of charge results in weaker binding, necessitating the presence of additional components to balance the positive charge of the metal ions. A mixed-ligand approach, combining carboxylates and pyridines, can therefore be employed to simultaneously satisfy coordination and charge requirements. Beyond these, polyazole molecules such as imidazole and pyrazole offer distinct advantages, as they can deprotonate to form anionic, multiterminal ligands with strong basicity. This enables them to establish robust coordination with metal ions, imparting enhanced stability to the resulting MOFs. Consequently, MOFs synthesized from common metal centers and diverse organic ligands have demonstrated significant potential as drug carriers, owing to their structural versatility and functional adaptability.

1.1.2 MOFs based on Zn^{2+} :

When considering MOFs as drug carriers, one of the primary concerns is their potential toxicity and the extent to which it may affect biocompatibility. It is therefore essential that both the metal ions and organic ligands used in constructing drug-delivery MOFs exhibit favorable biocompatibility profiles. Certain metals, such as chromium and nickel, are highly toxic and unsuitable for biomedical applications, whereas others—including iron, zinc, copper, and manganese—are essential trace elements naturally present in the human body at relatively high concentrations (e.g., plasma iron at ~22 mM, tissue zinc at ~180 μM , copper at ~68 μM , and manganese at ~180 μM). These non-toxic, endogenous metal ions are thus preferred for the synthesis of drug-carrying MOFs. Among them, Zn^{2+} has been widely employed due to its safety and versatility, with several synthetic approaches available, including room-temperature solution reactions, solvothermal methods, electrodeposition–solvothermal techniques, and microfluidic synthesis. Furthermore, post-synthetic modification of Zn-MOFs can significantly enhance their drug loading capacity, sustain release profiles, and enable controlled release mechanisms. Multifunctional Zn-MOFs can also be rationally designed to integrate therapeutic and diagnostic functions, for example by incorporating imaging agents for theranostics or photothermal conversion agents to achieve synergistic effects between chemotherapy and photothermal therapy [1].

1.2 ZIFs:

Zeolitic imidazolate frameworks (ZIFs) represent a distinctive subclass of reticular structures derived from tetrahedral, four-connected networks analogous to those found in zeolites and minerals. Constructed from transition-metal ions coordinated with imidazolate-type linkers, ZIFs offer precise control over pore size, shape, surface area, and functionality. Reticular chemistry, which links molecular building units through strong bonds to form extended crystalline architectures, has given rise to several important framework materials, including MOFs, COFs, and ZIFs. ZIFs are particularly notable for their zeolite-like structures, in which the Si–O–Si fragment of zeolites is conceptually replaced by M–L–M linkages, where M denotes tetrahedral metal ions such as Zn, Co, Cd, Cu, or Fe, and L represents an imidazolate linker. The preferred bond angle around oxygen in zeolites ($\sim 145^\circ$) closely matches that of bridging imidazolate linkers, enabling the formation of more than 100 distinct ZIF structures. Beyond reproducing known zeolite topologies, ZIFs have expanded the structural landscape by introducing new tetrahedral frameworks. Synthetic strategies such as mixed-linker approaches have further broadened the accessible structure space, allowing the creation of ZIFs with large apertures, cavities, and novel topologies. These features highlight the unique pore metrics and linker flexibility of ZIFs, which underpin their exceptional chemical and thermal stability. Moreover, the use of small imidazolate linkers enables the construction of frameworks with unusually large pore openings and cavities. The intrinsically reticulated three-dimensional cage-like structures of ZIFs provide a solid-state alternative to molecular cages, overcoming limitations associated with Brownian motion in solution and allowing enhanced control over chemical confinement within pores without solvent interference. Zeolitic imidazolate frameworks (ZIFs), a subclass of MOFs with zeolite-like structures, are typically synthesized by combining tetrahedral metal ions such as Zn, Cd, and Co with imidazolate ligands. These frameworks are distinguished by their exceptionally high surface areas, unimodal micropores, and remarkable chemical and thermal stability. Structurally, ZIFs possess minute pore openings of approximately 3.4 Å and cavities of about 11.6 Å, which restrict the entry of larger molecules and thereby enhance selectivity. Their robustness is further demonstrated by their ability to withstand thermal conditions up to 550 °C under a nitrogen atmosphere. Moreover, ZIFs exhibit outstanding chemical resistance, showing no structural degradation even after prolonged incubation in water and organic solvents at 50 °C for seven days. These unique properties—high porosity, stability, and selective permeability—make ZIFs highly attractive for applications in catalysis, adsorption, and drug delivery [2].

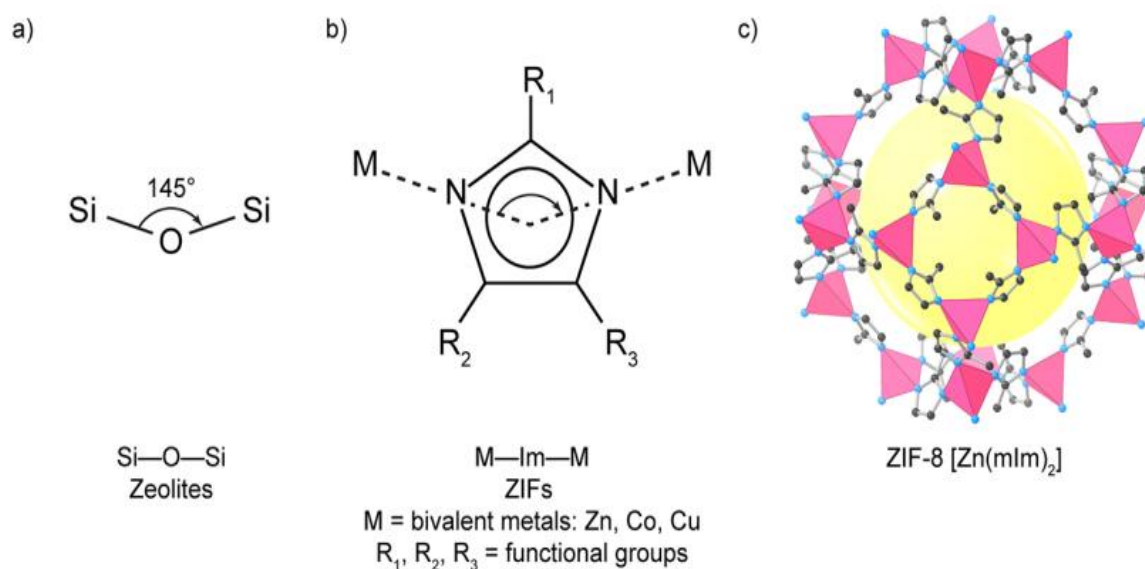


Figure 1.2 (a) Bridging angle (145°) of Si–O–Si in zeolites. (b) The M–Im–M in ZIFs, which is a requisite for synthesizing zeolite-type structures, as presented in ZIF-8 structure [Zn (mIm)₂] (mIm = 2-methylimidazolate). (c) Crystal structure of ZIF-8 viewed from the outside of the *t*-*toc* cage. Colour code: Zn, pink; C, black; N, blue. H atoms are omitted for clarity. [3]

The properties, types, and structural diversity of zeolitic imidazolate frameworks (ZIFs) are largely determined by the specific combinations of imidazole-based linkers and transition-metal ions employed during synthesis. A wide range of ZIF structures has been reported, typically constructed using metals such as Zn and Co, linkers including 2-methylimidazole and benzimidazole, and solvents such as water, dimethylformamide (DMF), diethylformamide (DEF), ethanol, and methanol. Conventional synthetic approaches for ZIFs include solvothermal and hydrothermal methods, generally carried out at temperatures between 298–423 K and often requiring an excess of imidazole to drive framework formation. While these methods have successfully expanded the structural landscape of ZIFs, ongoing research emphasizes the development of more efficient synthetic strategies aimed at improving chemical yield and minimizing environmental impact. Such advancements are critical for enabling sustainable production and broadening the practical applicability of ZIFs in catalysis, adsorption, sensing, and biomedical fields [4].

1.2.1 Types of ZIF:

ZIF – 7:

Zeolitic imidazolate framework ZIF-7 ($\text{Zn}(\text{bim})_2$) is one of the prototypical members of the ZIF family, formed by the coordination of zinc ions with benzimidazole (bim) linkers. Single-crystal X-ray diffraction studies reveal that ZIF-7 adopts a sodalite (SOD) topology with a hexagonal rhombohedral space group (R3), distinguishing it structurally from ZIF-8, which crystallizes in a cubic space group. This difference in crystal symmetry results in distinct physicochemical properties between the two frameworks. Compared to ZIF-8, ZIF-7 possesses smaller pore apertures, yet it demonstrates a gate-opening phenomenon that enables the selective adsorption of molecules such as ethylbenzene (EB), m-xylene (MX), and o-xylene (OX). ZIF-7 is therefore considered a promising microcrystalline material with unique adsorption behavior. Its gas uptake properties, particularly for hydrogen, have been studied both theoretically and experimentally. Although hydrogen is a renewable and gravimetrically dense fuel, the limited storage capacity of ZIF-7 remains a challenge, preventing it from achieving competitiveness with conventional fuels such as gasoline. Nonetheless, the structural distinctiveness and selective adsorption capabilities of ZIF-7 highlight its potential for further development in gas storage and separation applications [5].

ZIF – 67:

ZIF-67 has emerged as one of the most extensively utilized metal–organic frameworks, particularly valued as a soft template for diverse applications. Its intrinsic properties—including high catalytic activity, tunable pore size, and exceptional chemical and thermal stability—make it an attractive candidate for both fundamental research and large-scale applications. Importantly, the performance of ZIF-67 can be further enhanced by integrating it with other components or structures, resulting in hybrid materials with superior functionalities compared to the pristine framework. Among these, metal oxide nanoparticle/ZIF-67 composites represent a rapidly developing class of materials that combine the tailored porosity and structural versatility of ZIF-67 with the functional diversity of metal oxide crystalline phases. A wide range of such composites has been synthesized and systematically evaluated, demonstrating promising performance in adsorption, catalysis, sensing, energy storage, and microwave absorption. These studies highlight not only the critical role of ZIF-67-derived metal oxides in expanding application domains but also the

importance of synthesis strategies and morphological control in optimizing their functional properties [6].

ZIF – L:

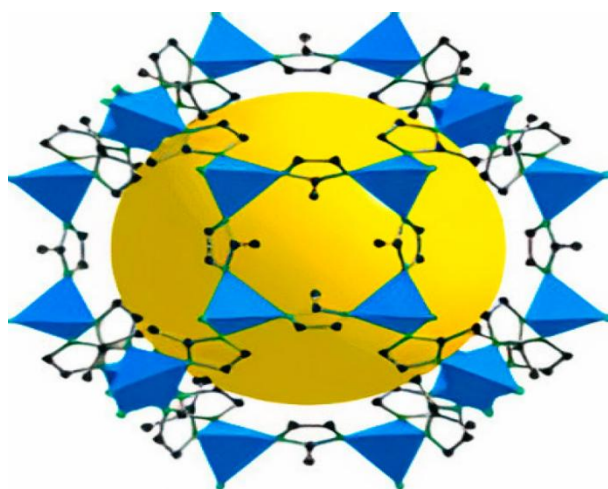
Zeolitic imidazolate framework ZIF-L is a distinctive MOF material synthesized via a simple aqueous solution route, yielding a two-dimensional leaf-like morphology. ZIF-L exhibits pronounced antibacterial activity under both simulated physiological conditions and marine environments. Its nanosheet structure demonstrates complete inactivation of *Escherichia coli* within 4 hours in phosphate-buffered saline (PBS) and *Bacillus subtilis* within 30 minutes in seawater, underscoring its potent antimicrobial efficacy. Furthermore, ZIF-L incorporated into epoxy composites enhances functional performance, providing not only strong antibacterial effects but also antifouling and algicidal properties against diverse marine algae. Remarkably, the ZIF-L/epoxy composite achieves a removal rate of 90.20% for *E. coli* with the addition of only 0.25 wt% ZIF-L, highlighting its efficiency at low loading levels. These findings emphasize the significant potential of ZIF-L and its composites in antibacterial and anti-algae applications, positioning metal–organic materials as promising candidates for biomedical and marine environmental technologies [7].

ZIF- 8:

ZIF-8 is a porous polymeric framework composed of zinc metal centres tetrahedrally coordinated to 2-methylimidazole ligands, forming a sodalite (SOD) topology with a remarkable surface area of approximately 1600 m² g⁻¹ and a six-membered ring pore aperture of 3.4 Å. This material exhibits exceptional thermal stability, retaining its structural integrity up to 400 °C. Density Functional Theory (DFT) studies reveal that the external surface of ZIF-8 hosts Lewis acidic and basic sites along with Brønsted acidic sites, whereas such active sites are absent within the internal pore environment. At the nanoscale, ZIF-8 crystals (nZIF-8) can be conveniently synthesized under ambient conditions in methanol, where the crystal formation proceeds through three distinct stages: nucleation, progressive crystal growth, and a stationary phase marking the completion of the process. [8].

ZIF-8, with its well-defined porous framework, demonstrates the ability to encapsulate small bioactive molecules such as antibiotics and secondary metabolites from medicinal plant extracts, thereby serving as an efficient platform for targeted drug delivery. Encapsulation studies have successfully incorporated therapeutic agents including doxorubicin (DOX), vancomycin, gentamicin, and ciprofloxacin within its pores. In addition to its role as a carrier,

ZIF-8 and its derived compounds exhibit intrinsic antibacterial activity, attributed to the presence of zinc ions and the porous architecture. The release of zinc ions induces rupture and deformation of bacterial cell walls, while simultaneously generating an alkaline microenvironment that inhibits microbial proliferation. This dual functionality is particularly significant in the context of antibiotic resistance, a global challenge driven by the overuse and misuse of conventional antibiotics, which enables bacteria to adapt, multiply, and render drugs ineffective. As resistance continues to rise among both gram-positive and gram-negative pathogens, alternative treatment strategies are urgently required. In this regard, ZIF-8 and its derivatives emerge as promising candidates, offering a synergistic approach that combines drug encapsulation with inherent antibacterial properties to combat drug-resistant microorganisms [2]. The unique structural characteristics of ZIF-8 impart remarkable flexibility and multifunctionality, enabling its integration into a wide spectrum of advanced applications. Recent studies highlight its effectiveness in lithium–sulfur batteries, where the porous framework enhances sulfur confinement and electrochemical performance. Similarly, ZIF-8 has been employed in electrocatalysis, offering active sites and stability for energy conversion reactions. Beyond energy systems, its biocompatibility and tunable porosity make it an excellent nanocomposite scaffold for wound healing, while its pH-responsive behavior has been exploited in controlled drug delivery systems. In the field of photovoltaics, ZIF-8 contributes to the development of efficient perovskite solar cells, and its adsorption capacity has been harnessed for water treatment applications. Collectively, these diverse utilizations underscore the versatility of ZIF-8 as a multifunctional material bridging energy, biomedical, and environmental domains [9].



ZIF-8

Figure 1.2.1 Structure of ZIF – 8

1.3 Biomedical applications of ZIF-8-based materials:

Antibacterial properties:

ZIF-8 has attracted considerable interest in recent years owing to its large pore size and versatile functionality, particularly within the biomedical domain. One of its most notable roles is as an antimicrobial agent, where it effectively eliminates bacteria through the release of metal ions from its framework or dopants, directly damaging bacterial cell walls. This ion release initiates a cascade of antibacterial mechanisms, including the generation of reactive oxygen species (ROS), disruption of internal organelles, and degradation of glutathione (GSH). Several studies have explored ZIF-8-based materials for antibacterial applications, with promising outcomes. For instance, ZIF-8 modified with functionalized graphene oxide (GO) containing amine groups via direct post-modification in ammonium hydroxide solution was incorporated into a polyethersulfone (PES) matrix to form a composite film. This composite demonstrated significant antibacterial activity, achieving inhibition rates of 81.1% against *Escherichia coli* and 85.7% against *Staphylococcus aureus*. In another approach, ZIF-8 synthesized by conventional methods was subjected to heat treatment in an oxygen environment to produce thermally sensitive ZIF-8 (T-ZIF-8). Remarkably, T-ZIF-8 exhibited a bactericidal efficiency of 99.99% against both *E. coli* and *S. aureus* within just 20 minutes, far surpassing the performance of the ZIF-8/GO composite. This superior activity was attributed to its photocatalytic properties, which enhanced ROS generation, particularly radicals such as $O_2^{\cdot-}$ and $OH^{\cdot}$, thereby accelerating bacterial destruction. Collectively, these findings underscore the potential of ZIF-8 and its derivatives as powerful antimicrobial platforms, offering innovative solutions to combat bacterial infections [11].

Anticancer properties:

ZIF-8 has emerged as a highly versatile framework owing to its high porosity, structural regularity, and excellent modifiability, with recent studies highlighting its notable anticancer potential. The anticancer activity of ZIF-8 is primarily attributed to the release of intrinsic Zn^{2+} ions or doped metal ions (M^{n+}), which stimulate the generation of reactive oxygen species (ROS) to induce cell death. In particular, dopants such as Cu^{2+} and Fe^{2+} can penetrate the intracellular environment and participate in Fenton reactions, thereby producing hydroxyl

radicals ($\cdot\text{OH}$) that accelerate cellular destruction. The efficacy of ZIF-8-based composites has been well-documented through innovative designs. For example, nanosized ZIF-8/Au/CuS/folic acid composites synthesized via a self-template method demonstrated enhanced chemodynamic therapy (CDT), where CuS facilitated the catalytic decomposition of H_2O_2 into $\cdot\text{OH}$ radicals, effectively reducing HepG2 cancer cell viability to approximately 25%. In another approach, a supramolecular $\text{O}_2\text{-Cu/ZIF-8@ZIF-8@WP6}$ -methylene blue photosensitive system was developed for photodynamic therapy (PDT) against HepG2 cells. This system achieved remarkable results, with only 5% of HepG2 cells surviving post-treatment, significantly outperforming $\text{O}_2\text{-Cu/ZIF@Ce6/ZIF-8@F127}$. The integration of ZIF-8 with WP6-methylene blue and PDT thus exemplifies the synergistic enhancement of anticancer efficacy, underscoring the potential of ZIF-8-based materials as multifunctional platforms in advanced cancer therapy.

Biosensing platform:

The development of ultra-sensitive and highly selective biosensors has emerged as a crucial strategy for early disease diagnosis and monitoring therapeutic outcomes. Among the various materials explored, ZIF-8-based biosensors have gained significant attention due to their tunable pore structures and versatile surface chemistry, which make them ideal platforms for immobilizing enzymes, antibodies, and aptamers in electrochemical sensing systems. By modifying electrode surfaces with ZIF-8 composites, two primary detection pathways—antibody immobilization and aptamer immobilization—can be effectively employed, thereby enabling highly specific and sensitive recognition of disease-related biomarkers, microRNAs, and even live cancer cells. Several studies have demonstrated the potential of ZIF-8-based biosensors in diverse biomedical applications. For instance, a multifunctional biosensor integrating ZIF-8 with graphene quantum dots was designed for the detection of M.SssI methyltransferases, enzymes that catalyze aberrant DNA methylation and are closely linked to disease progression. Remarkably, this biosensor exhibited a broad linear response range from 0.005 to 150 U mL^{-1} and achieved an ultra-low detection limit of 0.004 U mL^{-1} , underscoring its exceptional sensitivity and selectivity. Such findings highlight the promise of ZIF-8-based biosensors as powerful tools for early diagnosis and timely therapeutic intervention, paving the way for innovative biosensing therapies with significant clinical impact

Drug delivery system:

ZIF-8-based nanomaterials have attracted considerable attention in drug delivery research owing to their intrinsic porous architecture, high drug-loading capacity, pH-sensitive release behavior, and excellent thermal and chemical stability. These unique attributes render ZIF-8 an outstanding candidate for advanced drug delivery systems, particularly in cancer therapy where the acidic tumor microenvironment can be exploited for targeted release. The pH-responsive nature of ZIF-8 is equally advantageous in oral drug delivery, allowing therapeutic agents to bypass the harsh gastric environment and achieve efficient absorption in the alkaline milieu of the small intestine. Several studies have demonstrated the adaptability of ZIF-8 in such systems. For example, a ZIF-8-based nanocomposite grown on poly(3-caprolactone)-block-poly(quaternized vinylbenzyl chloride/bipyridine) polymers was employed to encapsulate doxorubicin hydrochloride (DOX) within micellar cores and internal cavities, with ZIF-8 serving as a protective matrix. This system achieved a release of nearly 80% DOX at pH 5.5 compared to only 10% at pH 7.4, confirming its selective responsiveness. Similarly, developed a cis-aconitic anhydride-doxorubicin (CAD) prodrug anchored within ZIF-8 and further functionalized with folic acid (FA) to enhance cancer-targeting efficiency. The release mechanism was highly controlled, proceeding through three sequential stages:

- (i) FA dissociation under acidic conditions, structural disruption of ZIF-8, and cleavage of the pH-sensitive linker.
- (ii) This design enabled minimal drug leakage in normal cells while ensuring efficient release in cancer cells, with more than 95% DOX released at pH 5.5 and negligible release at pH 7.4.
- (iii) Collectively, these findings underscore the potential of ZIF-8-based nanomaterials as intelligent, pH-responsive carriers that open new avenues for precise and effective drug delivery to targeted sites [11].

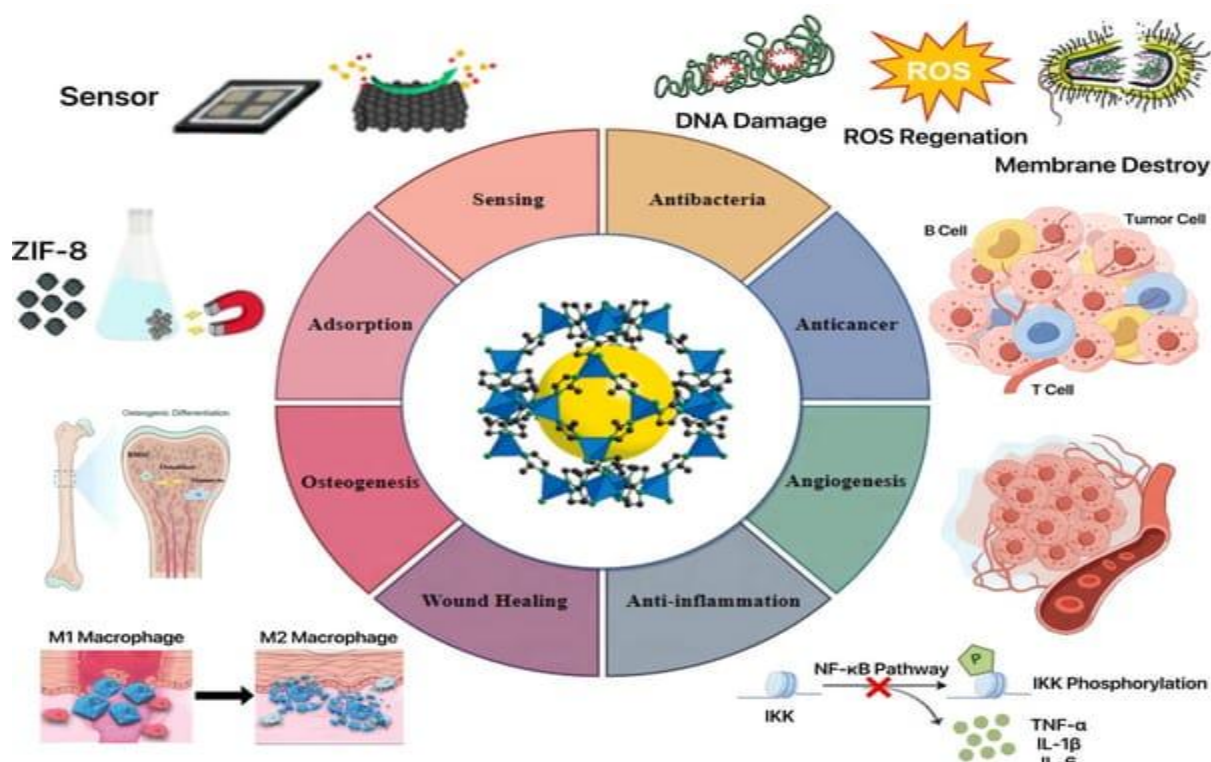


Figure 1.3 Biomedical applications of ZIF-8 [29]

1.4 PLANT USED:

AZADIRACHTA INDICA:

Neem (*Azadirachta indica*) has gained global recognition due to its extensive range of medicinal properties. The leaves and their bioactive constituents have been shown to possess immunomodulatory, anti-inflammatory, antihyperglycaemic, antiulcer, antimalarial, antifungal, antibacterial, antioxidant, antimutagenic, and anticarcinogenic activities, making neem a versatile therapeutic resource. Chemically, neem elaborates a diverse spectrum of biologically active compounds, with more than 140 molecules isolated from different parts of the tree. Among these, quercetin and β -sitosterol were the first polyphenolic flavonoids purified from fresh neem leaves, both exhibiting notable antibacterial and antifungal effects. High-performance liquid chromatography (HPLC) analysis of neem organic extracts further revealed the presence of major compounds such as 6-deacetylnimbin, azadiradione, nimbin, salannin, and epoxy-azadiradione, all of which demonstrated appreciable bioactivity against pathogenic fungi. The chemopreventive potential of neem leaf extract has also been validated in murine carcinogenesis models, where administration to Swiss albino mice significantly

reduced tumor incidence. Collectively, these findings establish that the Indian neem tree contains at least 35 biologically active principles, underscoring its remarkable therapeutic potential and positioning it as a promising candidate for the development of novel preventive and curative strategies [12]. Neem (*Azadirachta indica*) has long been recognized as an excellent wound-healing agent due to its potent antiseptic and regenerative properties. Its ability to enhance vascular permeability facilitates increased blood flow to the affected site, thereby accelerating the healing process. Moreover, neem promotes the rapid synthesis of collagen fibers, which are essential for closing wounds and restoring tissue integrity. In addition to its role in treating minor cuts and abrasions, neem has also been reported to be effective in managing skin burns, further underscoring its therapeutic versatility. Beyond wound healing, neem contains antipyretic compounds that have traditionally been employed to reduce fevers, highlighting its multifaceted medicinal potential. Collectively, these attributes position neem as a valuable natural resource in both traditional and modern therapeutic applications, reinforcing its significance in integrative medicine and preventive healthcare [13].



Figure 3. *Azadirachta indica* leaves

ACALYPHA INDICA:

Acalypha indica Linn. (Family: Euphorbiaceae), commonly known as Kuppimeni in Tamil, Kuppichettu or Harita Manjiri in Telugu, and Kuppigida in Kannada, is a small annual shrub that typically grows as a weed in gardens, roadsides, and across the plains of India. Despite its reputation as a troublesome plant, it has long been valued in traditional medicine for its diverse therapeutic applications. The leaves are known to possess laxative and antiparasitic properties, and when ground with common salt, quicklime, or lime juice, they are applied

externally to treat scabies. Similarly, a paste of the leaves with lime juice is prescribed for ringworm, while the fresh leaf juice serves as an emetic for children. Decoctions of the leaves are administered for earache, and powdered dry leaves are given to children to expel intestinal worms, often combined with garlic for enhanced efficacy. In homeopathy, *A. indica* has been employed in the treatment of severe cough accompanied by haemoptysis, bleeding from the lungs, and incipient phthisis. Phytochemical investigations have revealed that the plant contains a wide array of bioactive compounds, including kaempferol, a cyanogenetic glucoside, triacetoneamine, and the alkaloid acalphyne. Additional constituents such as acalphyamide and related amides, 2-methylanthraquinone, tri-O-methyl ellagic acid, γ -sitosterol, β -sitosterol, β -sitosterol glucoside, stigmasterol, n-octacosanol, quinine, tannins, resins, and essential oils further underscore its pharmacological richness. Collectively, these findings highlight *Acalypha indica* as a plant of considerable medicinal importance, bridging traditional knowledge with modern phytochemical validation [30]. *Acalypha indica* L., a weed widely distributed across the plains of India, has been traditionally recognized for its diverse medicinal applications. It has been reported to be effective in the treatment of pneumonia, asthma, rheumatism, and several other ailments. The dried leaves are commonly prepared as a poultice for bedsores and wounds, while the fresh juice, when mixed with oil or lime, is applied to alleviate a variety of skin disorders. Interestingly, the leaves of *Acalypha grandis* have also been associated with contraceptive activity, further expanding the therapeutic relevance of this genus. Beyond its direct medicinal uses, *A. indica* is of particular interest in the context of mosquito-borne diseases. Mosquitoes, especially *Anopheles stephensi* Liston, serve as primary vectors of malaria in India and West Asia, and their bites can trigger allergic responses ranging from localized skin reactions to systemic manifestations such as urticaria and angioedema. In this regard, botanical and microbial insecticides have gained prominence due to their efficacy and safety for non-target organisms. Moreover, the development of insect growth regulators has provided selective control strategies against agricultural, medical, and veterinary insect pests, often inducing mortality through neurotoxic effects. Phytomedicinally, *Acalypha indica* is recognized for its broad therapeutic potential, functioning as an emetic, expectorant, laxative, and diuretic. These properties render the plant particularly valuable in the management of respiratory and pulmonary disorders such as bronchitis, pneumonia, asthma, and tuberculosis. Taken together, the diverse pharmacological activities and ecological relevance of *A. indica* underscore its multifaceted significance. By bridging traditional ethnomedicinal practices with modern approaches to disease management and vector control, this plant exemplifies the integration

of indigenous knowledge with contemporary biomedical research, thereby reinforcing its importance as both a therapeutic agent and a subject of scientific inquiry.



Figure 4. Acalypha indica leaves

1.4 Extraction methods for studying phytochemicals:

Extraction from plants is essentially an empirical process, as different solvents are employed under varying conditions of time and temperature to optimize the yield of bioactive components. Once extracted, it becomes crucial to separate these active constituents from co-extractive compounds to ensure purity and efficacy. This is typically achieved through further fractionation, which is carried out on the basis of parameters such as acidity, polarity, or molecular size. Such systematic fractionation not only enhances the isolation of specific phytochemicals but also facilitates their subsequent characterization and application. The commonly employed extraction methods, along with their principles and relevance, are discussed in the following section, providing a comprehensive understanding of how solvent choice and operational conditions influence the recovery of plant-derived bioactive compounds.

1.4.1 Cold extraction method:

Plant extraction is essentially carried out by drying the different parts of the plant under controlled artificial conditions at relatively low temperatures (50–60 °C). The dried material is then powdered and used for solvent extraction. A measured quantity of the powdered

sample is transferred into a conical flask containing the respective solvent, and the mixture is maintained at room temperature for seven days, with intermittent shaking every twenty-four hours to facilitate efficient extraction. At the end of the extraction period, the mixture is filtered under vacuum using Whatman filter paper, and the filtrate is subsequently dried at room temperature in a watch glass dish. The weight of each dish is recorded both before and after drying, and the final yield of the extract is calculated from the difference in weight. This systematic approach ensures reproducibility and accuracy in quantifying the extracted bioactive compounds, thereby providing a reliable basis for subsequent phytochemical analysis and biological evaluation.

1.4.2 Solvent extraction method:

The Universal Extraction System (Buchi) has recently been adopted as a reliable method for solvent extraction of plant materials. In this procedure, the dried powder of various plant parts is carefully placed in a glass thimble and subjected to extraction using appropriate solvents. The process is typically carried out for ten cycles for each extract, with the temperature adjusted to remain just below the boiling point of the respective solvent to ensure efficient recovery of bioactive compounds without degradation. Following extraction, the resulting solvent fractions are filtered and concentrated using a vacuum concentrator, after which they are employed for the qualitative and quantitative determination of phytoconstituents. This systematic approach not only enhances reproducibility but also ensures the integrity of sensitive plant-derived compounds, thereby providing a robust foundation for phytochemical analysis and subsequent biological evaluation.

1.4.3 Supercritical fluid extraction (SFE):

Supercritical Fluid Extraction (SFE) represents an advanced technique that utilizes gases, most commonly carbon dioxide (CO₂), which are compressed into a dense liquid phase to act as the extracting solvent. In this process, the liquefied CO₂ is pumped through a cylinder containing the plant material, allowing the bioactive compounds to dissolve into the fluid. The extract-laden liquid is then directed into a separation chamber, where the desired phytoconstituents are efficiently separated from the gas. Importantly, the CO₂ is recovered and recycled for subsequent use, making the method both economical and environmentally sustainable. The solvent properties of CO₂ can be finely tuned by adjusting parameters such as pressure and temperature, thereby enabling selective extraction of specific compounds. A key advantage of SFE lies in its ability to leave no solvent residues, as CO₂ evaporates

completely, ensuring the purity of the extract. This makes SFE a highly reliable and clean approach for isolating sensitive plant-derived bioactive molecules, reinforcing its significance in modern phytochemical and pharmaceutical research.

1.4.4 Microwave-assisted extraction (MAE):

It is simply termed as microwave extraction, that combines microwave and traditional solvent extraction. Heating the solvents and plant tissue using microwave increases the kinetic of extraction, is called microwave-assisted extraction [6]. The target for heating in dried plant material is the minute microscopic traces of moisture that occurs in plant cells. The heating up of this moisture inside the plant cell due to microwave effect, results in evaporation and generates tremendous pressure on the cell wall. The cell wall is pushed from inside due to the pressure and the cell wall ruptures. Thus the exudation of active constituents from the ruptured cells occurs, hence increasing the yield of phytoconstituents [15].

1.4.5 Ultrasonic-Assisted Extraction:

Ultrasound-assisted extraction (UAE) has emerged as a versatile technique in food-processing and phytochemical research for isolating bioactive compounds from plant materials. Operating at frequencies above 20 kHz, ultrasound effectively disrupts plant cell walls, thereby enhancing solvent penetration and improving extraction yield. A key advantage of UAE lies in its ability to function at relatively low operating temperatures, which helps preserve the integrity and quality of sensitive compounds throughout processing. The method is considered one of the simplest extraction techniques, as it can be performed using common laboratory equipment such as an ultrasonic bath. In practice, a finely ground sample is mixed with the appropriate solvent and placed in the ultrasonic bath, with both temperature and extraction time carefully controlled to optimize recovery. UAE accommodates a wide range of solvents and can be applied to both organic and inorganic samples. Equipment typically includes ultrasonic baths and ultrasonic probe systems, although the latter presents limitations due to issues with experimental repeatability and reproducibility. Overall, UAE offers a rapid, efficient, and accessible approach to plant extraction, reinforcing its importance in modern phytochemical and food-processing applications. [16].

1.4.6 Soxhlet extraction:

Soxhlet extraction has been a standard technique for over a century and the methods based on it remain the primary references against which performance in new leaching methods is measured. The advantages and shortcomings of Soxhlet extraction have been used as starting points for the development of a variety of modifications intended to alleviate or suppress the latter while keeping or even improving the former. Most of the modifications reported over the last few decades have been aimed at bringing Soxhlet closer to that of the more recent techniques for solid sample preparation, by shortening leaching times with the use of auxiliary energies and automating the extraction assembly.

a) Conventional Soxhlet extraction:

In its classical application, Soxhlet extraction was originally developed for the determination of fat content in milk and has since become a widely adopted technique for solid–liquid extraction. In this method, the sample is placed in a thimble holder, which is gradually filled with condensed fresh extractant from a distillation flask. Once the liquid reaches the overflow level, a siphon mechanism transfers the solute-containing extract back into the distillation flask, thereby carrying the analytes into the bulk solvent. This cycle is repeated continuously until the extraction process is complete. Operationally, Soxhlet extraction functions as a continuous–discrete system: while the extractant acts in a stepwise manner, the recirculation of solvent through the sample imparts a continuous character to the process. Conventional Soxhlet extraction offers several notable advantages. The repeated contact of the sample with fresh portions of solvent facilitates displacement of the transfer equilibrium, thereby improving extraction efficiency. Additionally, the system operates at relatively high temperatures due to the heat applied to the distillation flask, which partially extends to the extraction cavity. Another practical benefit is that no filtration is required after leaching, and sample throughput can be increased by performing multiple extractions in parallel, a feature made feasible by the relatively low cost of the basic equipment. Collectively, these attributes underscore the enduring relevance of Soxhlet extraction as a robust and accessible technique, while also providing the foundation for modern modifications aimed at enhancing efficiency and reproducibility [17].

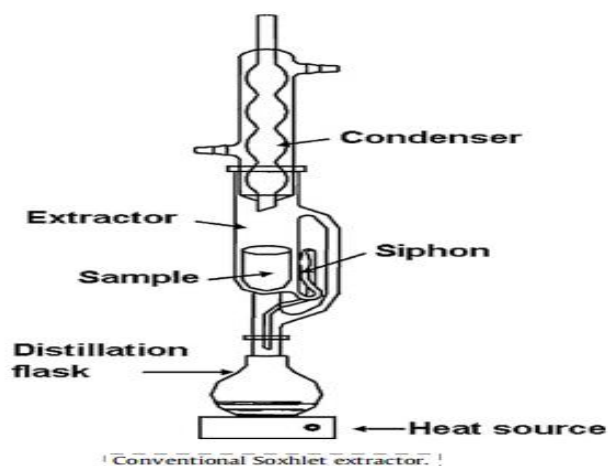


Figure 1.4.6 Conventional Soxhlet extractor

b) Ultrasound-assisted Soxhlet extraction:

An extractor based on the classical physico-chemical principles of Soxhlet, but enhanced through the application of ultrasound, was designed, constructed, and successfully applied by the authors' team for the extraction of total fat from oil seeds such as sunflower, rapeseed, and soybean. This innovative approach employs conventional Soxhlet glassware, with the Soxhlet chamber placed in a thermostatic bath where ultrasound is delivered via an ultrasonic probe. The introduction of ultrasound energy into the sample cartridge disrupts the matrix and accelerates solvent penetration, thereby improving extraction efficiency. As a result, the method yields outcomes comparable to, and in some cases superior to, those obtained using conventional Soxhlet leaching, which is recognized as the official ISO reference method. This integration of ultrasound with Soxhlet extraction demonstrates how traditional techniques can be modernized to achieve higher performance, reduced extraction times, and enhanced reproducibility, thereby reinforcing the relevance of Soxhlet-based systems in contemporary analytical and phytochemical research.

c) Microwave-assisted Soxhlet extraction:

Microwave-assisted Soxhlet extraction has emerged as one of the most significant advancements in improving the performance of conventional Soxhlet methodology, offering a wider range of approaches compared to other modifications. This technique integrates microwave irradiation with the classical Soxhlet process, introducing several distinctive features. First, the extraction vessel remains open, ensuring that the procedure operates under normal atmospheric pressure. Second, microwave energy is specifically directed toward the sample compartment, thereby enhancing the efficiency of solvent-sample interaction. Third, the extraction step is carried out

either partially or entirely in the conventional Soxhlet manner, maintaining continuous contact between the sample and fresh portions of extractant. Finally, the process eliminates the need for subsequent filtration, simplifying the workflow. Collectively, these attributes make microwave-assisted Soxhlet extraction a particularly compelling improvement, as it combines the robustness of the classical technique with the speed and efficiency of microwave energy, thereby reinforcing its relevance in modern phytochemical and analytical applications. [17].

CHAPTER – 2

2.1 REVIEW OF LITERATURE

Atul Arya *et al.*, (2023) reported DC@ZIF-8 nanocomposites revealed outstanding antibacterial efficacy against the cephalosporin-carbapenem resistant *Escherichia coli* (CCREC), methicillin-resistant *Staphylococcus aureus* (MRSA), chloramphenicol-resistant *Bacillus subtilis* (MTCC-441) and β -lactams-resistant *Pseudomonas aeruginosa* (MTCC-1688) strains. DC@ZIF-8 has better electrostatic interaction with the bacterial membrane in comparison to bare ZIF-8 framework and plant extract that results into greater zone of inhibition. Antibacterial compounds present in *Drymaria cordata* extract and enhanced bio efficacy of DC@ZIF-8 nanomaterials against pathogenic bacteria. DC@ZIF-8 nanocomposite has endless therapeutic potential to act as a targeted drug delivery agent against bacterial disorders.

Parisa Ghazanfari *et al.*, (2025) reported *Pterocarya fraxinifolia* leaf extract (PF) into a zeolitic imidazolate framework-8 (ZIF-8), aiming to enhance its stability and antimicrobial efficacy. The antimicrobial activity of PF@ZIF-8 was evaluated against *Candida albicans*, *Aspergillus flavus*, *Staphylococcus aureus*, and *Escherichia coli* using well and disk diffusion methods. Compared to the free extract and ZIF-8 alone, the PF@ZIF-8 nanocomposite-particularly at 20% extract loading-demonstrated significantly larger inhibition zones and lower minimum inhibitory concentration (MIC) values. These results suggest a synergistic interaction between the plant bio actives and the ZIF-8 carrier, leading to improved efficacy. Although cyto-toxicity assays were not performed, existing literature supports the potential

safety of ZIF-8 in biomedical applications, especially when used in combination with natural agents.

Lakshmanan Govindan *et al.*, (2025) fabricated a bioactive *Achyranthes aspera* leaf extract loaded with ZIF-8 nanocomposite for improved wound healing and anticancer applications. This nanocomposite has the potential to release phytochemicals and zinc ions in the targeted cell sites due to the intracellular caspase-3 and apoptotic mechanism. The results of biocompatibility studies suggested, there were no other toxic effects observed against the *Artemia salina* animal model. The anticancer activity results suggested that AA@ZIF-8 leads to nuclear dysfunction and ROS mediated cytotoxicity against A549 cells. The bioactive potential of AA@ZIF-8 nanocomposite causes excellent cytotoxic efficacy in lower concentrations. The overall experimental findings strongly indicated that the AA@ ZIF-8 nanocomposite is a potential biomaterial for anticancer applications.

Qiuxiang Wang *et al.*, (2020) review introduces the preparation process, functional modification methods of ZIF-8, and the latest research progress of ZIF-8 composite materials, and introduces the application of ZIF-8 in drug delivery and tumor therapy in detail. ZIF-8 has pH response characteristics, so it can release high concentrations of drugs after reaching the acidic environment of the tumor. In addition, ZIF-8 can also be designed as other stimulus-responsive drug delivery systems (light, heat, magnetic field, targeted location, GSH, H₂O₂ stimulation, etc.) to control the release of drugs at specific sites to reduce the toxic side effect on normal tissue sites. Composite materials with multiple functions are currently a hot spot in the field of biomedical research. For example, nano-gold and graphene can be composited with ZIF-8 to prepare multifunctional biomaterials. These multifunctional materials can improve the drug loading, stability, and biocompatibility of ZIF-8 drug delivery system, realizing integrated diagnosis and treatment, and multi-therapy synergistic treatment of cancer.

Zhiling Zheng *et al.*, (2023) reported the synthesis of wide range of ZIFs. Then analysed the crystal structure of ZIFs and relationship between their structure and properties of ZIF. ZIFs are a fascinating class of porous crystals because of their reticulated cage structures and tunable properties. The ability to diversify ZIF structures through the manipulation of linker chemistry and topologies has allowed for the creation of a wide range of crystalline solids. The linker-linker interaction and its use in the context of mixed-linkers impact the size of the

pore opening and internal diameter and provide a powerful guide to accessing progressively larger pore ZIFs.

Thuan Van Tran *et al.*, (2025) investigated the recent developments in ZIF-8 synthesis techniques and critically discuss innovative approaches such as the use of green solvents and advanced methods such as microwave- and ultrasound-assisted syntheses. The multifunctional applications of ZIF-8-based biomaterials in biomedical engineering are critically explored with their pivotal roles in antibacterial and anticancer therapies, drug delivery systems, bone tissue engineering, and diagnostic platforms such as biosensing and bioimaging. The present review also clarifies some innovations of ZIF-8-based materials in pH-sensitive and glucose-responsive drug delivery systems and scaffolds for bone regeneration.

Vahid Hoseinpour *et al.*, (2021) investigated the application of the ZIF-8 in bone tissue engineering. It has been found that the Ti implants coated with nanoscale ZIF-8 could develop extracellular matrix mineralization, upregulate the expression of osteogenesis related genes (Alp and Runx2) in MG63 cells, promote alkaline phosphatase (ALP) activity, and inhibit the growth of *Streptococcus mutans* compared to the pure and/or porous Ti. As a result, ZIF-8 NPs coatings were recognized as potentially beneficial materials for the surface modification of implants. Additionally, ZIF-8 could be employed in biodegradable scaffolds and/or it could be fabricated as a separation layer on the porous inorganic support.

Fatma Şahin *et al.*, (2018) reported the ZIF-8, ZIF-L, ZIF-7 and ZIF-67 were synthesized from fresh and recycled synthesis solutions. ZIF-8, ZIF-L and ZIF-7 includes Zn^{2+} , while ZIF-67 includes Co^{2+} . ZIF-7 was synthesized in dimethyl formamide, while the others in water. from fresh and recycled synthesis solutions with high purity and yield. The adjustment of pH and the metal ion to organic linker molar ratio of the mother liquor were crucial for the synthesis of next generation of the ZIF crystals through the recycling procedure. The crystals produced from recycled solutions had similar morphological and structural properties with those produced from fresh synthesis solutions. Those ZIFs represent synthesis using different metal ions, organic linkers and solvents.

Meisam Malekmohammadi *et al.*, (2019) reported that solvent has a direct effect on phase purity, crystallinity and crystal size distribution. Pure ZIF-8 phase was obtained from aqueous synthesis solution only at 25 °C and ligand/metal molar ratio of 4 while synthesized samples from methanol solution with stoichiometric ligand/metal molar ratio of 2 at both high and

room temperatures were pure highly crystalline ZIF-8. The samples synthesized at different conditions owned versatile textural properties. ZIF-8 yielded from methanol solution at high temperature possessed more uniform particles with higher degree of crystallinity compared to ZIF-8 obtained from aqueous solution at room temperature. The phase thermal stability of the samples were similar with only a minor difference in the weight loss of ZIF-8 sample prepared in water-based system in the temperature range of 100–200 °C.

Zhao Wei *et al.*, (2025) reported BC-based multifunctional sponge for diabetic wound dressing using a simple three-step process: in situ growth of Nar-encapsulated ZIF-8, ethanol treatment, and lyophilization. The incorporation of Nar/ZIF-8 NPs into BC notably improved the physicochemical and biological properties of the composite sponge. The Nar/ZIF-8/BC sponges are intact, flexible, and low density (8.6 g/cm³), making them easy to transport, store, and clinical practice. Due to the hydrogen bonding between ZIF-8 and the hydroxyl groups on BC fibers, the mechanical properties of the Nar/ ZIF-8/BC sponge have been significantly enhanced, with a tensile strength reaching up to 2.28 MPa. The incorporation of ZIF-8 provides broad-spectrum antimicrobial activity and enables drug release in response to changes in the DFU microenvironment. Together, the Nar/ZIF-8/BC multifunctional sponge effectively promotes diabetic-infected wound healing in rats by regulating the microenvironment through exudate management, antimicrobial, antioxidant, angiogenic, and anti-inflammatory effects.

Erfan Behjat *et al.*, (2025) fabricated a series of multifunctional nanocomposite hydrogels by incorporating ZIF-8@TA nanoparticles and fresh garlic extract into a PVA-borax matrix. Among those, the optimized formulation (PBTZ10/Garlic) demonstrated a unique combination of pH-responsive behavior, self-healing capacity, robust mechanical strength, and enhanced biological activity. The hydrogels exhibited tunable swelling behavior, reaching up to 184% in acidic and 260% in basic environments, which enabled accelerated degradation and controlled release of therapeutic ions. The garlic extract, rich in sulfur-containing bioactive compounds, not only enhanced the antioxidant and antibacterial effects but also contributed to the hydrogel's overall biocompatibility, as evidenced by favorable cellular responses. These findings confirm that the developed nanocomposite hydrogels, especially PBTZ10/Garlic possess a balanced integration of structural, physicochemical, and biological features.

U. Seeta Uthaya Kumar *et al.*, (2020) reported the seaweed-based films incorporated with neem leaves extract and treated with gamma radiation. The films showed substantial improvement in the mechanical with excellent antimicrobial activity towards *Staphylococcus aureus* and *Bacillus subtilis*, which was highly desirable for the packaging application. It indicates that the irradiation doses up to 2.5 kGy on the biocomposite films showed reduced water vapour permeability, moisture content, and water solubility percentages. The seaweed-neem films also showed an increase in film transparency up to a radiation dose of 2.5 kGy as consumers desire transparency, and it presented a good visual presentation for the packaged product. The morphology of the irradiated composite films displayed a smoother surface compared to the seaweed-neem film (control). It resulted in the enhanced mechanical strength of the films with irradiation.

Yasodani Nishshanka *et al.*, (2025) reported that neem–chitosan composite was synthesized to serve as an effective drug endowed with antioxidant properties, potential to inhibit protein denaturation, and antibacterial properties, operating as a controlled/sustained drug delivery system. This was achieved by integrating ethanolic *Azadirachta indica* leaf extract with chitosan biopolymer, both of which were derived from natural sources. The neem leaf extract, rich in bioactive compounds such as phenols, tannins, flavonoids, glycosides, steroids, and terpenoids, contributed to its enhanced medicinal properties. Meanwhile, chitosan served as an exceptional drug carrier. Characterization of the composite through scanning electron microscopy (SEM) revealed the crystallization of neem extract on the porous structure of chitosan. The antibacterial effect of the composite resulted from a synergistic interaction between the neem leaf extract and chitosan, displaying enhanced activity against Gram-positive and Gram-negative bacteria through both bacteriostatic and bactericidal mechanisms.

Nagarajan Sakthiguru *et al.*, (2021) investigated biocomposite composed of Chitosan (Cs), Zinc oxide (ZnO), *Acalypha indica* (AI) was fabricated by simple chemical precipitation method. The anti-inflammatory and antioxidant studies were studied using Diclofenac sodium and ascorbic acid as standard by Fetal bovine serum denaturation and (2,2-diphenyl-1-picryl-hydrazyl-hydrate) DPPH radical scavenging method. The effective bactericidal action of the composite was evaluated against both gram-negative, gram-positive bacteria such as *Escherichia coli* (E.coli) and *Staphylococcus aureus* (S.aureus). The hemolysis activity was studied using Triton x 100 as negative and Phosphate buffer solution (PBS) as positive control. The invitro wound healing potential was also assessed by scratch wound

assay with L929 mouse fibroblast cells. Obtained results suggest that the Cs/ZnO/AI biocomposite can act as a suitable candidate in wound care applications.

Aswathy Priyamvadan *et al.*, (2023) observed that chitosan, polyvinyl alcohol, copper oxide and *Acalypha indica* (AI) extract have proven antimicrobial efficacy either individually or in combination. The presented study was focused on development of chitosan/PVA/CuO-AI nanocomposite for antimicrobial activity. In addition, hemolytic activity of the nanocomposite was also determined. In addition, the chitosan/ PVA/CuO-AI nanocomposite demonstrated effective antimicrobial activity and it was not hemolytic on human blood samples. In vitro cytotoxicity demonstrated that the viability of the *Danio rerio* gill (DrG) cell treated with various doses of CS/PVA/CuO-AI was shown to be greater than 85% at different concentrations and no significant morphological alterations were seen. The results from the study demonstrated that although the prepared CS/PVA/CuO-AI shows good antibacterial activity.

Su, Yadi, Junqi Wang *et al.*, (2025) reported the biomedical domain has witnessed a significant incorporation of nanotechnology in recent years. Zeolitic imidazolate framework-8 (ZIF-8) has demonstrated potential as a pH-responsive smart targeted release nanoplatfrom for antibacterial and anti-inflammatory agents, attributed to its superior biocompatibility and robust physicochemical stability. Furthermore, the high porosity and substantial surface area of ZIF-8 enable significant drug-loading capacity and the encapsulation of various agents, including small inorganic molecules and large protein molecules. Moreover, ZIF-8 composite materials exhibit unique pH sensitivity and photosensitivity, resulting in decomposition and subsequent release of antibacterial and anti-inflammatory agents under acidic conditions and near-infrared irradiation. To synthesize recent advancements in pH-responsive and photosensitive ZIF-8 as a dual-responsive smart targeted release nanoplatfrom in antibacterial therapy and wound regeneration.

De Villenoisy *et al.*, (2023) reported Materials derived from metal–organic frameworks (MOFs) have demonstrated exceptional structural variety and complexity and can be synthesized using low-cost scalable methods. Although the inherent instability and low electrical conductivity of MOFs are largely responsible for their low uptake for catalysis and energy storage, a superior alternative is MOF-derived metal-based derivatives (MDs) as these can retain the complex nanostructures of MOFs while exhibiting stability and electrical conductivities of several orders of magnitude higher. The present work comprehensively

reviews MDs in terms of synthesis and their nanostructural design, including oxides, sulfides, phosphides, nitrides, carbides, transition metals, and other minor species. The aim of this approach is to provide an inclusive platform for the strategies to design and process these materials for specific applications.

RESEARCH GAP:

Metal-organic frameworks (MOFs), and particularly zeolitic imidazolate frameworks (ZIFs), have emerged as versatile platforms for applications ranging from catalysis and sensing to drug delivery and environmental remediation. Among them, ZIF-8 is especially attractive due to its high surface area, tunable pore dimensions, and remarkable chemical stability. Over the past decade, researchers have successfully demonstrated the encapsulation of conventional drugs, synthetic molecules, and a limited set of natural compounds—most notably curcumin, quercetin, and resveratrol—within ZIF-8 matrices. While these studies have advanced our understanding of MOF–biomolecule interactions, they represent only a narrow exploration of the vast phytochemical diversity available in medicinal plants.

Medicinal plants are reservoirs of complex phytochemical mixtures, including alkaloids, flavonoids, terpenoids, and phenolics, many of which act synergistically to produce

therapeutic effects. However, the majority of ZIF-8 studies have focused on purified, well-characterized compounds rather than whole or novel extracts. This leaves a significant gap: the potential of unexplored plant extracts as functional payloads within ZIF-8 systems has not been systematically investigated. Such extracts may exhibit unique chemical interactions with the MOF framework, influencing encapsulation efficiency, release kinetics, and even the stability of the carrier material.

Neem (*Azadirachta indica*) and kuppaimeni (*Acalypha indica*) are two medicinal plants with long histories of use in traditional medicine, valued for their antimicrobial, anti-inflammatory, and antioxidant properties. Despite their well-documented bioactivity, neither neem nor kuppaimeni extracts have been reported in the context of ZIF-8 encapsulation. This absence is striking, given their phytochemical richness and therapeutic relevance. Exploring these extracts within ZIF-8 systems not only broadens the chemical space studied but also provides an opportunity to evaluate how complex, synergistic mixtures behave in MOF-based delivery platforms. Such work could reveal new insights into bioavailability enhancement, controlled release mechanisms, and the stabilization of bioactive compounds.

SCOPE AND OBJECTIVES

- To prepare *azadirachta indica* plant extract.
- To prepare *acalypha indica* plant extract.
- To synthesise *azadirachta indica* loaded ZIF-8.
- To synthesise *acalypha indica* loaded ZIF-8.
- To characterise the synthesised plant extract loaded ZIF-8 using FT-IR, SEM, XRD, TGA, Antibacterial activities.

CHAPTER – 3

MATERIALS & METHODS:

Apparatus Required: Magnetic stirrer, pellet, beaker, burette, RB flask.

Materials Required: 2-methylimidazole was purchased from Sisco research laboratories Pt. Ltd Maharashtra, India. zinc nitrate hexahydrate was purchased from Sisco research laboratories Pt. Ltd Maharashtra, India. Methanol was supplied from Finar limited Gujarat, India.

3.1. Preparation of ZIF-8:

SUBSTANCE	WEIGHT/ml	MOLECULAR WEIGHT	EQUIVALENT WEIGHT	MOLE
2- METHYLIMIDAZOLE	4g	82.10 g/mol	82.11 g/eq	0.0487M
ZINC NITRATE HEXAHYDRATE	2g	297.49 g/mol	148.74 g/eq	0.00672M
METHANOL	60 ml	32.04 g/mol	32.04 g/eq	-

Table 3.1. Materials required for the preparation of ZIF- 8

2-methylimidazole (4 g) and zinc nitrate hexahydrate (2 g) were prepared in 40 mL and 20 mL of methanol, respectively. The 2-methylimidazole solution was stirred continuously with a magnetic stirrer at 350 rpm for 12 hours. After that, a solution of 2-methylimidazole was mixed with zinc nitrate hexahydrate using dripping. Then it allowed to settle down for 5 hours. Then centrifuge the solution and then washed the precipitate for three to four times with methanol. Then dried the precipitate in oven for more 12 hours.



Figure 3.1. Synthesis of ZIF-8

3.2. Preparation of plant extract:

Apparatus required: Soxhlet apparatus, RB flask, oil bath.

Materials required: Methanol was supplied from Finar limited Gujarat, India. Azadirachta indica leaves, Acalypha indica leaves are collected from Anna university CEG campus Chennai.

Procedure:

Collected Azadirachta indica leaves, Acalypha indica leaves is sun dried for 3 to 4 days. Grind the dried leaves to powder. 20g leaves powder was enclosed in cellulose filter paper

and placed inside a Soxhlet apparatus. The system was connected to a 250 mL round-bottom flask containing 150 mL of high-purity methanol. Extraction was carried out at a controlled temperature of 70 ± 2 °C for 48 h under reflux. Afterward, the extract was filtered through a filter paper to remove residual plant matter. Then the extract is stored in container under 4°C for further synthesis.

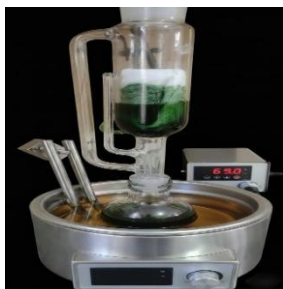


Figure 3.2. Soxhlet extraction of plants

3.3. Synthesis of Plant Extract-Encapsulated ZIF-8:

Apparatus Required: Magnetic stirrer, pellet, beaker, burette, RB flask.

Materials Required: 2-methylimidazole was purchased from Sisco research laboratories Pvt. Ltd Maharashtra, India. zinc nitrate hexahydrate was purchased from Sisco research laboratories Pvt. Ltd Maharashtra, India. Methanol was supplied from Finar limited Gujarat, India. Acalypha indica and Azadirachta indica plant extract.

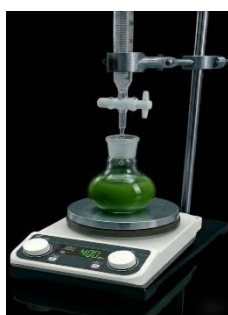
SUBSTANCE	WEIGHT/ml	MOLECULAR WEIGHT	EQUIVALENT WEIGHT	MOLE
2- METHYLIMIDAZOLE	4g	82.10 g/mol	82.11 g/eq	0.0487M
ZINC NITRATE HEXAHYDRATE	2g	297.49 g/mol	148.74 g/eq	0.00672M
METHANOL	60 ml	32.04 g/mol	32.04 g/eq	-
AZADIRACHTA	10 ml	-	-	-

INDICA				
ACALYPHA INDICA	10 ml	-	-	-

Table3.3. Materials required for Synthesis of Plant Extract-Encapsulated ZIF-8.

PROCEDURE:

To prepare the plant extract loaded (AZ@ZIF-8) & (AC@ZIF-8) 2.0 g of zinc nitrate hexahydrate was dissolved in 20mL of methanol. Separately, 4.0 g of 2-methylimidazole and 1.0 g of the methanolic extract were dissolved in 40mL of methanol and stirred at 400 rpm at room temperature for 24h. After the first hour of stirring, the zinc solution was gradually introduced dropwise (~1 drop/s) into the extract-containing solution over a period of 30min. A visible colour shift from cream to green colour indicated successful encapsulation. The resulting precipitate was recovered by centrifugation at 5000 rpm for 15min, washed three times with methanol, and oven-dried at 70 °C for 24h.



(a)



(b)

Figure 3.3 (a) & (b) synthesis of plant extract loaded ZIF-8

3.4 CHARACTERISATION:

INSTRUMENTATION DESCRIPTION:

3.4.1. THE FOURIER TRANSFORM INFRARED (FTIR)

The benefit of using infra-red spectroscopy is that it can examine almost any sample like Nano-material, films etc, in any state. Liquids, solutions, paste, films, powders, fibre, gases and surfaces can all be examined using the appropriate sampling technique. By identifying the changes in the functional groups, The Fourier transform infrared (FTIR) technique is sort of spectroscopy that can identify the changes in the overall composition of bio molecules is

used to measure the vibration and rotation of molecules affected by an infrared wavelength. Through the identification of structural vibrations in molecule binding, information about the existence of their interactions is ascertained.

3.4.2. X-RAY DIFFRACTION (XRD)

The benefit of using X-ray diffraction is that it can analyze crystalline materials such as nano-materials, powders, thin films, and bulk solids. It is widely used for the identification of crystal structure, phase composition, and other structural parameters. Samples like powders, crystals, thin films, and solid materials can be examined using appropriate sample preparation techniques. By identifying the diffraction patterns produced by the interaction of X-rays with the crystal lattice, X-ray diffraction (XRD) is a technique that can determine the arrangement of atoms in a material and analyze changes in its structural composition. It is used to measure the scattering of X-rays when they interact with the periodic arrangement of atoms in crystalline substances. Through the identification of diffraction peaks corresponding to specific lattice planes, information about crystal structure, phase purity, crystallite size, and lattice parameters can be ascertained. XRD also helps in understanding material properties such as strain, defects, and degree of crystallinity by analyzing peak positions and intensities.

3.4.3. THERMO GRAVIMETRIC ANALYSIS (TGA)

TGA is a technique to study the thermal behaviour of materials by measuring the change in mass of sample as it is heated in a controlled manner. Thermal stability of the membrane was studied by using TGA instrument PERKINELMER. During a typical TGA experiment, a small sample of the material is placed in a crucible and heated or cooled in a controlled manner while its weight is continuously monitored. The sample is typically heated at a constant rate, and the change in mass is recorded as a function of temperature or time. By analysing the weight loss or gain of the sample as it is heated or cooled, information about its thermal properties and composition can be obtained. TGA can provide information about a wide range of material properties, including Thermal stability, composition, and purity and decomposition kinetics.

3.4.4. ANTIBACTERIAL ACTIVITY

The benefit of antibacterial study is that it can evaluate the effectiveness of substances such as nano-materials, plant extracts, chemicals, and pharmaceutical compounds against bacterial strains. It is widely used in microbiology, medicine, and material science to determine the

ability of a substance to inhibit or kill bacteria. Samples like solutions, coatings, powders, and films can be tested using appropriate antibacterial assay techniques. By identifying the inhibition of bacterial growth, antibacterial study is a method that can determine the activity of compounds against microorganisms and analyze changes in bacterial viability. It is used to measure the interaction between antibacterial agents and bacterial cells, which affects their growth and survival. Through the observation of inhibition zones, reduction in colony formation, or decrease in turbidity, information about the antibacterial efficiency of the material is ascertained. Common methods include agar well diffusion, disc diffusion, and broth dilution techniques. These studies help in understanding the mechanism of action, effectiveness, and potential applications of antibacterial agents in healthcare and industrial fields.

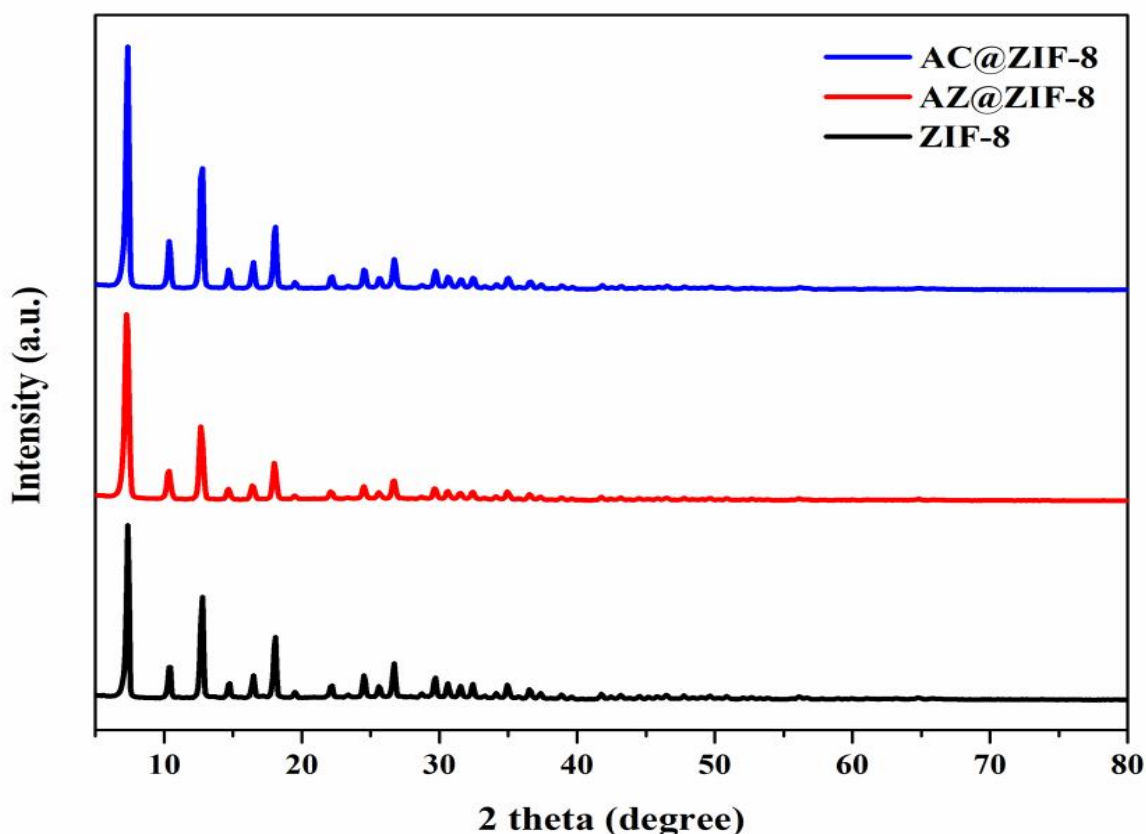
3.4.5. SCANNING ELECTRON MICROSCOPE (SEM)

SEM is a type of electron microscope that produces high-resolution images of a sample by scanning the surface with a focused beam of electrons. The electrons interact with atoms in the sample, producing various signals that contain information about the surface topography as well as cross section view. Surface morphology of the Nano-material was measured by using SEM instrument TESCAN (Model: VEGA 3XMU). SEM sample is typically coated with a thin layer of conductive material (gold) to produce electrically conductivity and prevent charging. Detectors in the SEM collected the backscattered or emitted electrons and use them to create an image of the surface of the sample. The resulting image provides

CHAPTER – 4

RESULTS AND DISCUSSIONS

4.1. X-RAY DIFFRACTION (XRD)



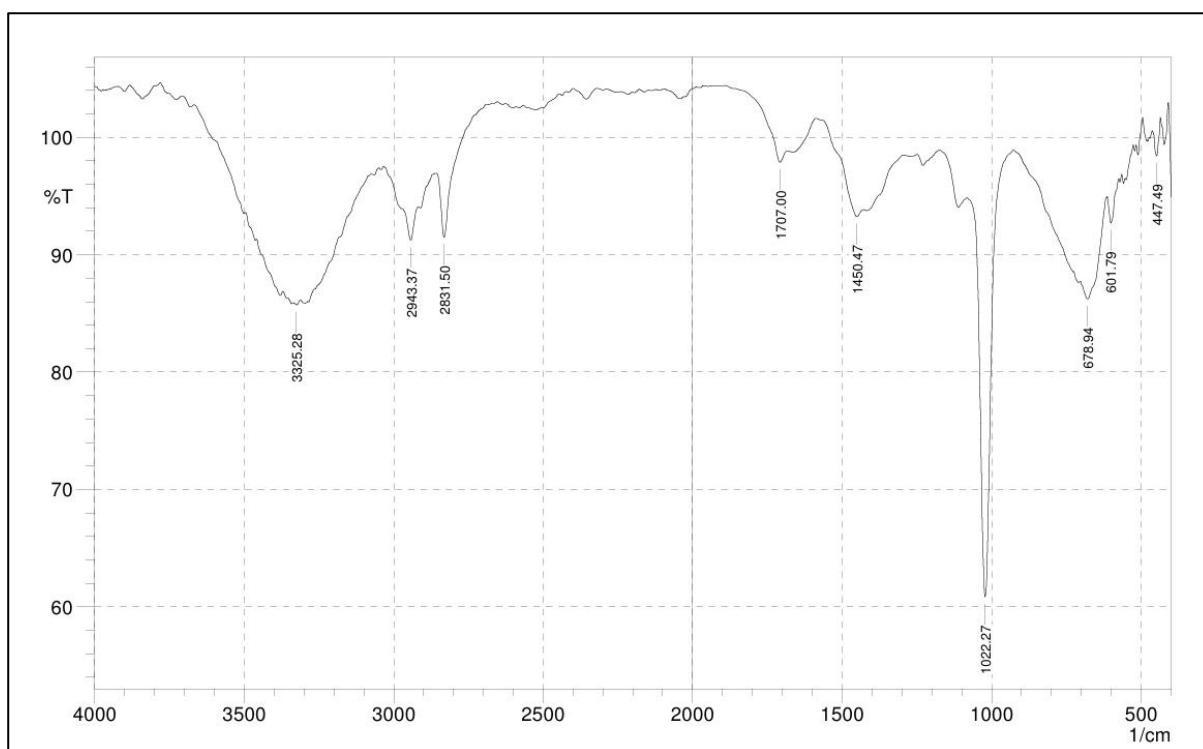
(A)

Figure (A) XRD of ZIF@8, AZ@ZIF-8, AC@ZIF-8

The X-ray diffraction patterns of pristine ZIF-8, neem extract-loaded ZIF-8 (AZ@ZIF-8), and Kuppaimenni extract-loaded ZIF-8 (AC@ZIF-8) are shown in the figure. The pure ZIF-8 sample exhibits sharp and intense diffraction peaks at 2θ values around 7.3° , 10.4° , 12.7° , 14.7° , 16.4° , and 18.0° , which are characteristic of the highly crystalline sodalite-type structure of ZIF-8. These peaks are in good agreement with the reported standard pattern, confirming the successful formation of crystalline ZIF-8 without any impurities. In the case of plant extract-loaded samples, both AZ@ZIF-8 (neem) and AC@ZIF-8 (Kuppaimenni) show similar diffraction patterns to that of pure ZIF-8. This indicates that the fundamental crystal structure of ZIF-8 is well preserved even after loading with plant extracts. However, a slight reduction in peak intensity is observed in the loaded samples compared to pristine ZIF-8. This decrease in intensity suggests partial coverage of the ZIF-8 surface and incorporation of bioactive compounds from the plant extracts into the pores or onto the surface of the framework. Additionally, minor peak broadening can be noticed in AZ@ZIF-8 and AC@ZIF-8, which may be attributed to a slight decrease in crystallinity due to the interaction

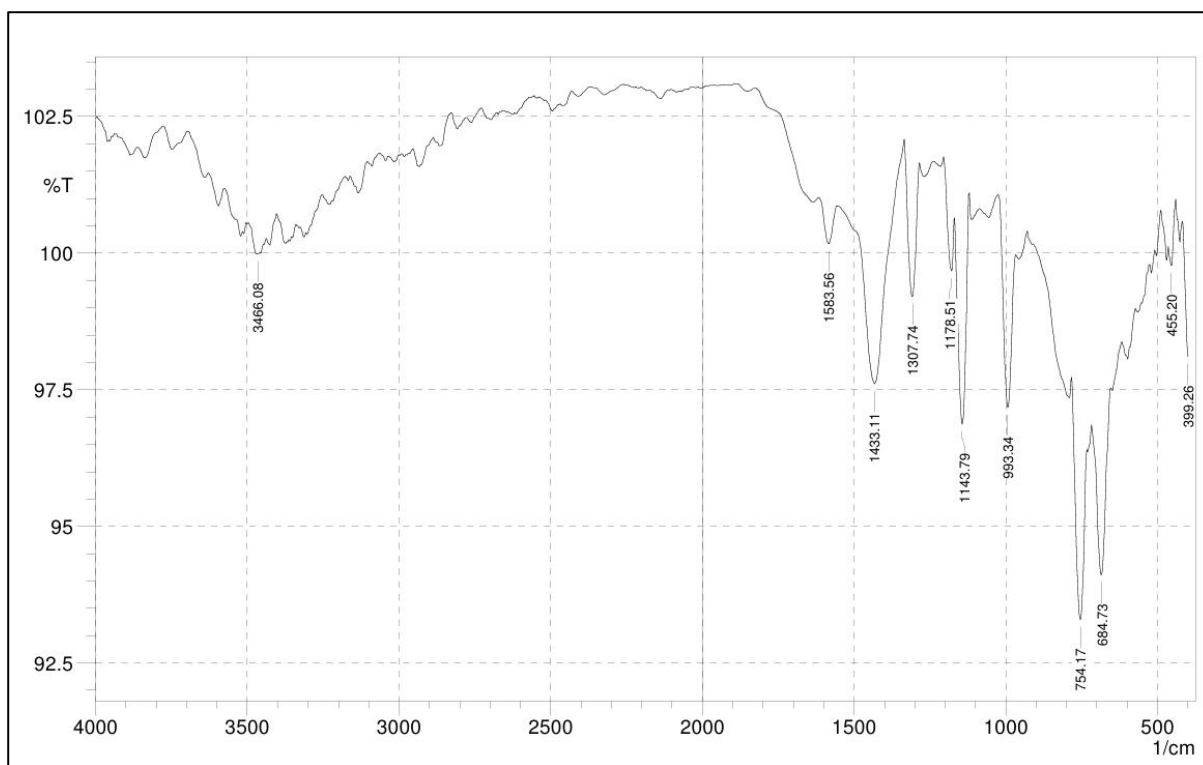
between ZIF-8 and phytochemicals present in neem and Kuppaimenni extracts. The absence of any new diffraction peaks in the loaded samples confirms that no separate crystalline phase is formed, and the plant extracts are successfully incorporated without altering the structural integrity of ZIF-8. Overall, the XRD results demonstrate that ZIF-8 maintains its crystalline framework after loading with plant extracts, while the slight changes in peak intensity and broadening confirm successful encapsulation of neem and Kuppaimenni bioactive compounds within the ZIF-8 structure.

4.2 FT-IR SPECTROSCOPY



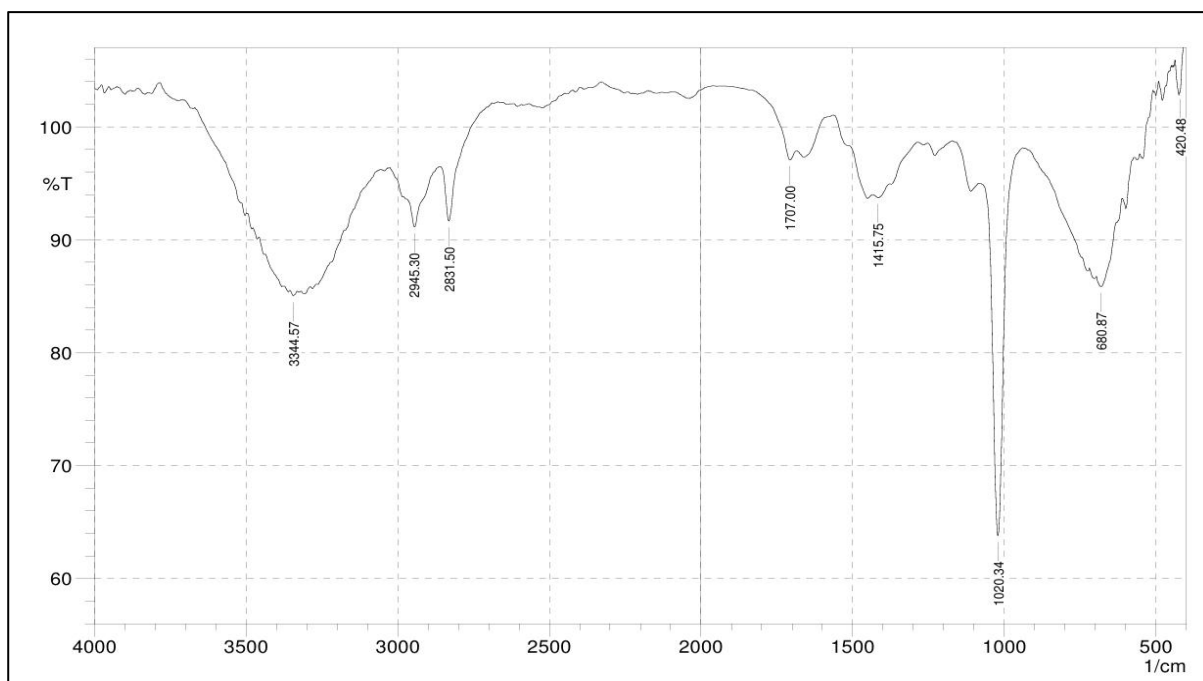
(a)

Figure (a) *Azadirachta indica* plant extract



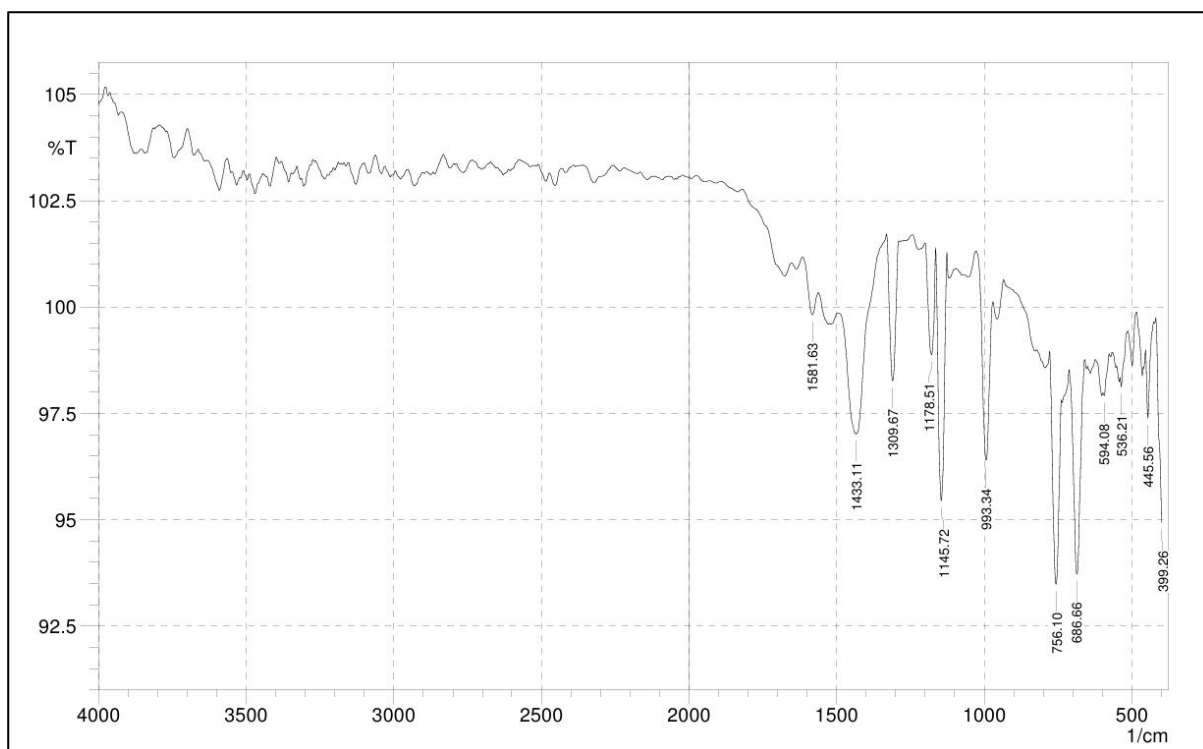
(b)

Figure (b) Azadirachta indica plant extract loaded ZIF-8 (AZ@ZIF-8)



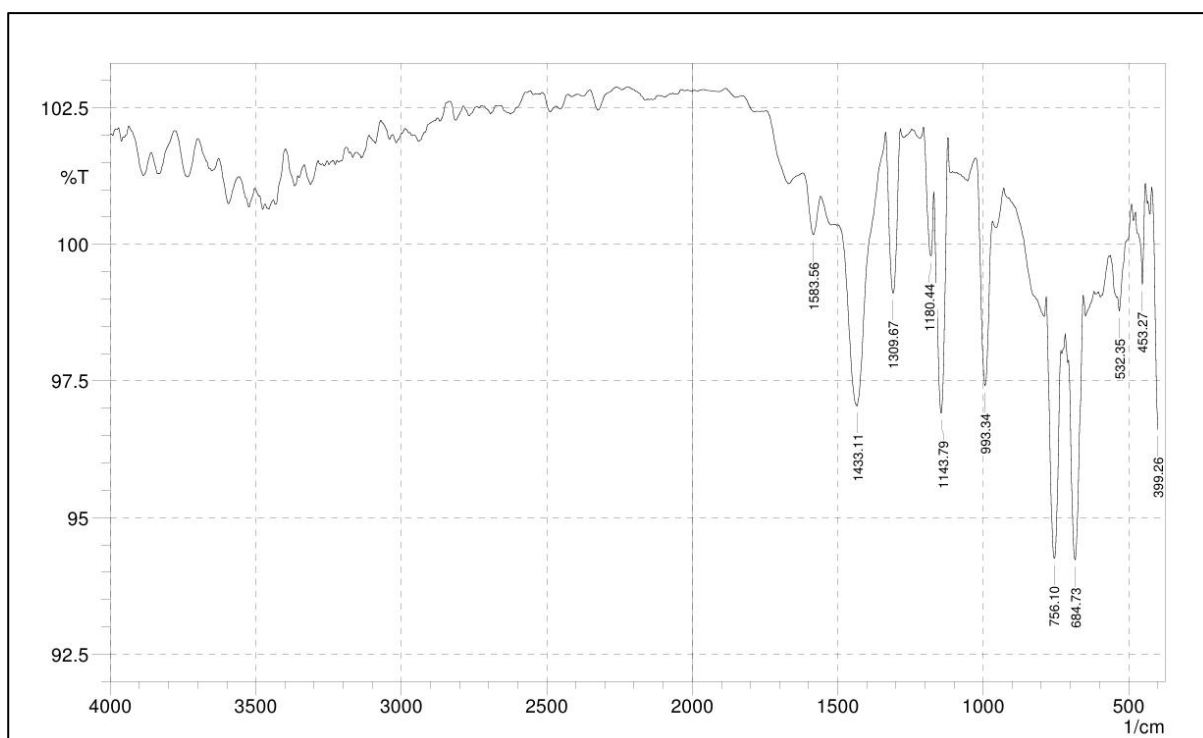
(C)

Figure (C) Acalypha indica plant extract



(d)

Figure (d) *Acalypha indica* plant extract loaded ZIF-8 (AC@ZIF-8)



(e)

Figure (e) ZIF-8

The FTIR spectra of both plant extracts (AZ and AC) exhibited characteristic absorption bands corresponding to diverse phytochemical functional groups. A broad band in the region of $\sim 3466\text{--}3344\text{ cm}^{-1}$ was attributed to O–H stretching vibrations, indicating the presence of hydroxyl groups associated with phenols and alcohols. The peaks observed at $\sim 2945\text{--}2831\text{ cm}^{-1}$ correspond to aliphatic C–H stretching of methylene groups. A prominent absorption band around $\sim 1707\text{ cm}^{-1}$ confirms the presence of C=O stretching vibrations, suggesting carbonyl-containing compounds such as aldehydes, ketones, or carboxylic acids. The bands at $\sim 1581\text{--}1450\text{ cm}^{-1}$ are assigned to C=C stretching vibrations, indicating aromatic or alkene structures, while peaks near $\sim 1433\text{ cm}^{-1}$ correspond to O–H bending or carboxylic functionalities. Further, absorption bands in the range of $\sim 1309\text{--}1020\text{ cm}^{-1}$ are attributed to C–N and C–O stretching vibrations, confirming the presence of amines, ethers, and esters. Peaks observed in the lower region ($\sim 756\text{--}601\text{ cm}^{-1}$) are associated with C-H bending vibrations of substituted aromatic rings, while signals below $\sim 600\text{ cm}^{-1}$ are related to skeletal vibrations. These findings confirm that both extracts are rich in bioactive compounds such as phenolics and flavonoids, which can act as reducing and stabilizing agents. In the FTIR spectra of AZ@ZIF-8 and AC@ZIF-8, the characteristic peaks of both the plant extracts and the ZIF-8 framework were observed with slight shifts and variations in intensity, confirming successful encapsulation. The O–H stretching bands ($\sim 3466\text{--}3344\text{ cm}^{-1}$) indicate the retention of hydroxyl groups after incorporation. The presence of peaks around $\sim 1583\text{ cm}^{-1}$ corresponds to C=N stretching vibrations of the imidazole ring, confirming the formation of the ZIF-8 structure. Bands at $\sim 1433\text{--}1415\text{ cm}^{-1}$ and $\sim 1307\text{--}1178\text{ cm}^{-1}$ are attributed to C–N stretching and in-plane bending of the imidazole ring. The strong absorption near $\sim 993\text{--}1020\text{ cm}^{-1}$ further supports C–N stretching vibrations within the framework. Additionally, peaks observed at $\sim 754\text{--}684\text{ cm}^{-1}$ correspond to aromatic C-H bending, while the distinct bands in the region of $\sim 455\text{--}399\text{ cm}^{-1}$ are assigned to Zn–N stretching, confirming metal-ligand coordination. Compared to the pure extracts, the observed shifts and intensity changes indicate strong interactions between phytochemical constituents and the ZIF-8 framework. The coexistence of extract-derived functional groups along with Zn–N vibrations confirms the successful loading of bioactive molecules into the ZIF-8 structure without disrupting its structural integrity.

SCANNING ELECTRON MICROSCOPE (SEM):

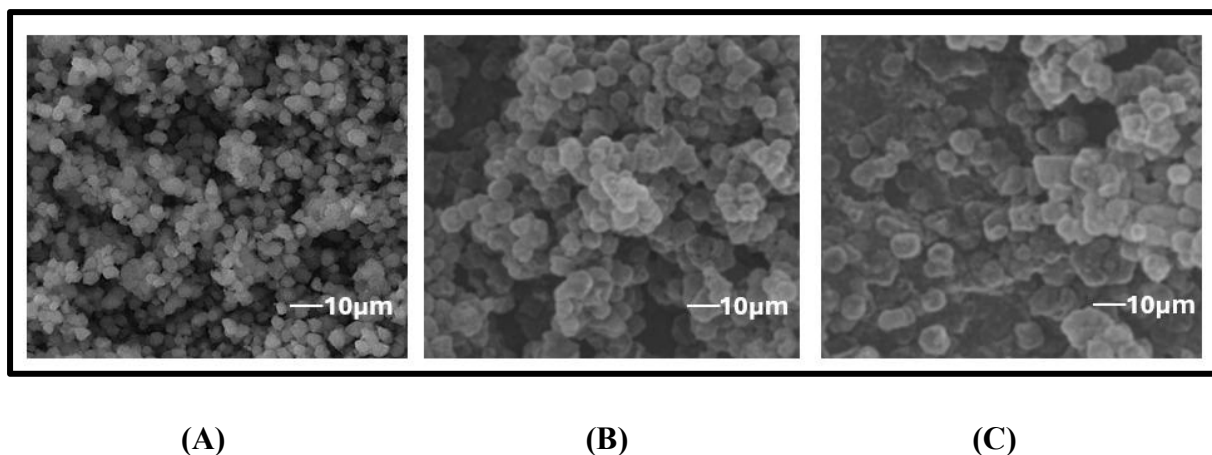
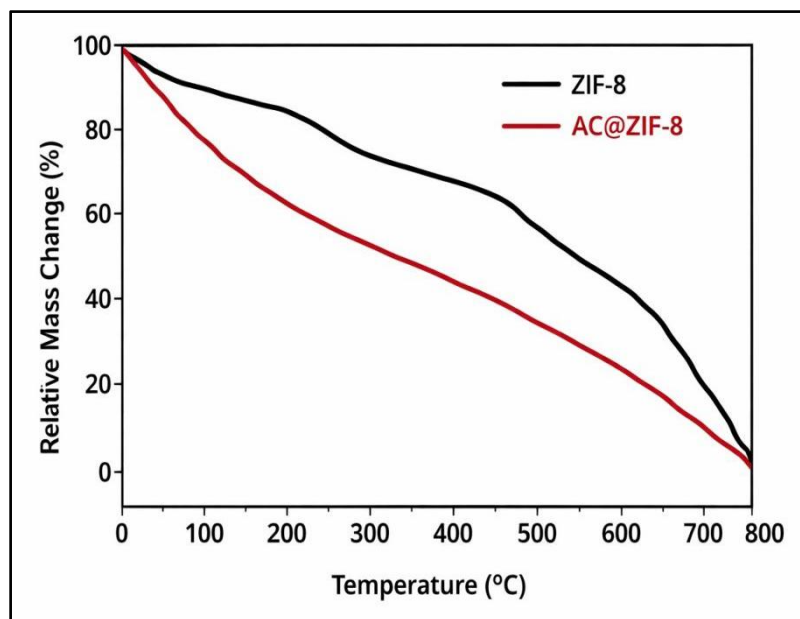


Figure (A) ZIF-8, (B) AZ@ZIF-8, (C) AC@ZIF-8

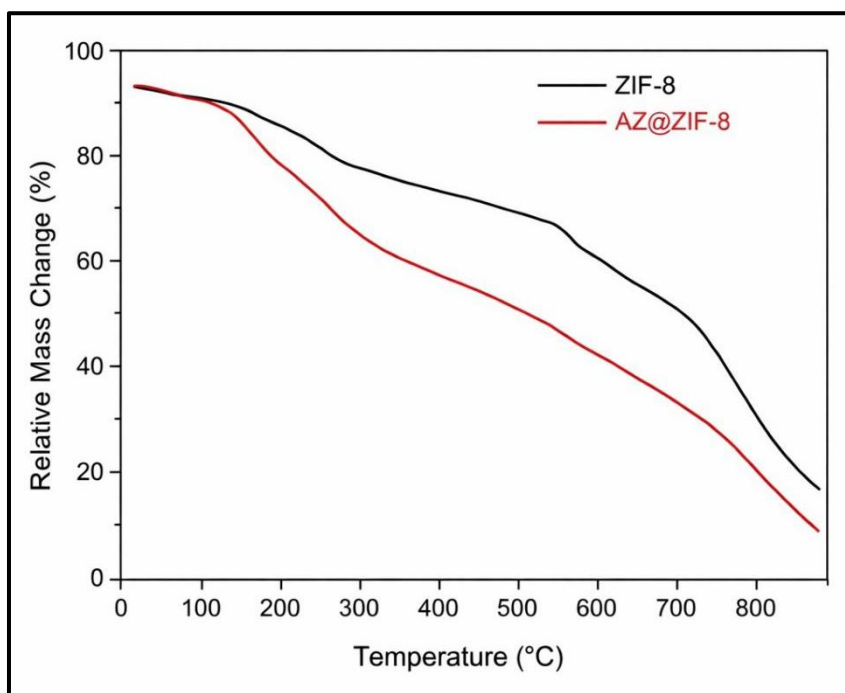
The SEM images show the surface morphology of ZIF-8 and plant extract loaded ZIF-8 samples. The pure ZIF-8 sample shows well-shaped particles with a uniform structure. Most of the particles appear as regular crystals with smooth surfaces, indicating good formation of ZIF-8. Slight agglomeration of particles is also observed. In the case of *Azadirachta indica*-loaded ZIF-8, the particles are not as uniform as pure ZIF-8. The surface appears covered with some irregular materials, which may be due to the presence of plant extract. This confirms that the extract is attached to the surface of ZIF-8. Similarly, the *Acalypha indica*-loaded ZIF-8 shows more clustering and rough surface. The particles are partially covered by additional deposits, indicating better interaction between the plant extract and ZIF-8. Overall, the SEM results suggest that the structure of ZIF-8 is maintained, but the surface is modified after loading with plant extracts. This confirms successful loading of the extracts.

THERMAL GRAVIMETRIC ANALYSIS (TGA) :



(A)

Thermal gravimetric analysis (TGA) was carried out to examine the mass variation of the samples under a nitrogen atmosphere (Fig. A). The results indicate that both ZIF-8 and AC@ZIF-8 remain stable below 100 °C. Over the 30–900 °C range, the two materials displayed distinct thermal behavior. ZIF-8 showed no appreciable mass loss until 200 °C, after which a sharp decline was observed. Between 200 and 250 °C, the framework exhibited a 10% reduction in mass. A cumulative loss of 23% was recorded up to 680 °C, corresponding to the progressive disintegration of the ZIF-8 structure. Complete degradation occurred between 800 and 900 °C, with nearly 80% mass loss due to collapse of the crystal lattice. Under oxidative conditions, the residue confirmed the conversion of ZIF-8 into zinc oxide. In contrast, AC@ZIF-8 demonstrated a more gradual mass loss profile. The onset of weight reduction after 200 °C is attributed to the depletion of encapsulated *Acalypha indica* extract. A minor 20% loss between 200 and 300 °C was associated with the removal of solvent molecules. A further 40% reduction occurred between 550 and 600 °C, corresponding to the decomposition of organic constituents derived from the plant extract. At higher temperatures (800–900 °C), an 80% mass loss was observed, indicating partial breakdown of the ZIF-8 framework and collapse of the crystalline structure. The overall thermal behavior confirms the successful encapsulation of *Acalypha indica* extract within ZIF-8.



(B)

Thermal gravimetric analysis was conducted under a nitrogen atmosphere to evaluate the mass variation of ZIF-8 and AZ@ZIF-8 (Fig. B). Both materials exhibited good stability below 100 °C, with negligible weight loss. Over the 30–850 °C range, distinct decomposition behaviors were observed. ZIF-8 remained intact up to ~200 °C, after which a noticeable decline in mass was recorded. Between 200 and 250 °C, the framework lost about 8–10% of its weight. A gradual reduction continued, reaching nearly 20% by 650 °C, which can be attributed to the progressive breakdown of the imidazolate framework. Complete degradation occurred between 800 and 850 °C, with ~75–80% mass loss, signifying collapse of the crystalline structure. The residue at higher temperatures confirmed the conversion of ZIF-8 into zinc oxide. AZ@ZIF-8, on the other hand, displayed a more continuous mass loss profile, reflecting the presence of encapsulated *Azadirachta indica* extract. The onset of weight reduction after 200 °C corresponded to the release of phytoconstituents. Between 200 and 300 °C, a ~15–20% mass loss was observed, associated with the removal of solvent molecules and volatile plant components. A further ~35–40% reduction occurred between 500 and 600 °C, linked to decomposition of organic constituents derived from the extract. At elevated temperatures (800–850 °C), an additional ~75–80% mass loss was recorded, indicating collapse of the ZIF-8 framework and disintegration of intermolecular bonds. These results confirm the successful encapsulation of *Azadirachta indica* extract within ZIF-8.

ANTI-BACTERIAL ACTIVITY

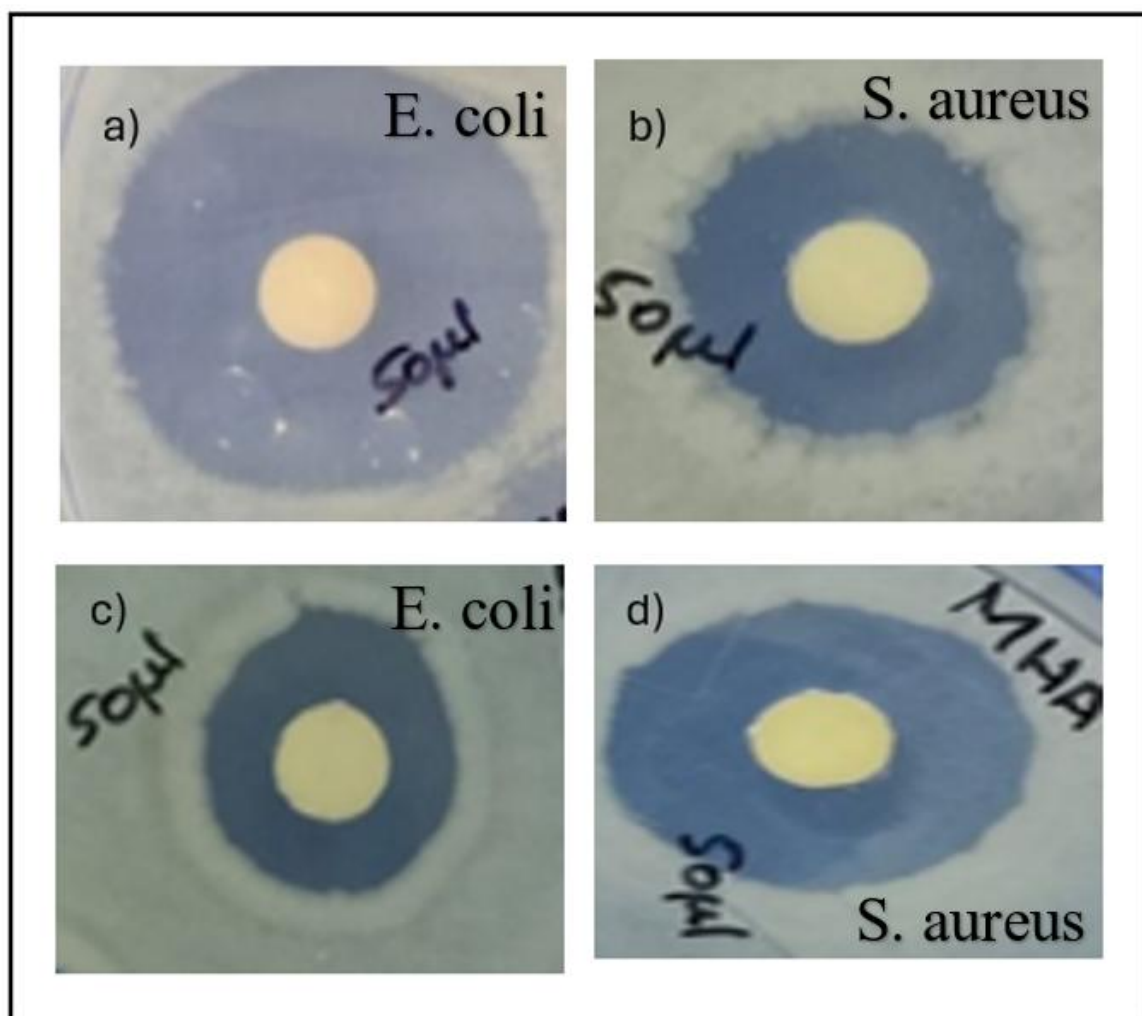


Figure. (a) and (b) *Azadirachta indica*, (c) and (d) *Acalypha indica*

The antibacterial activity of AZ@ZIF-8 (Neem) and AC@ZIF-8 (Kuppaimeni) was evaluated against *Escherichia coli* (Gram-negative) and *Staphylococcus aureus* (Gram-positive) using the agar diffusion method, and distinct zones of inhibition were observed in all cases. AZ@ZIF-8 exhibited the highest activity against *E. coli* with a zone of inhibition (ZOI) of 20 mm, whereas AC@ZIF-8 showed a lower inhibition of 14 mm against the same organism. In contrast, against *S. aureus*, both AZ@ZIF-8 and AC@ZIF-8 demonstrated comparable antibacterial efficacy, each producing a ZOI of 17 mm. The results clearly indicate that AZ@ZIF-8 has superior effectiveness, particularly against Gram-negative bacteria, while both composites are equally effective against Gram-positive bacteria. This variation in response can be attributed to differences in bacterial cell wall structure, where Gram-negative

E. coli possesses an outer membrane that typically restricts the penetration of antimicrobial agents, whereas Gram-positive *S. aureus* lacks this barrier, making it more susceptible. The strong activity of AZ@ZIF-8 against *E. coli* suggests that the neem-derived phytochemicals, in combination with Zn²⁺ ion release from the ZIF-8 framework, enhance permeability and induce membrane disruption and oxidative stress. The comparatively weaker activity of AC@ZIF-8 against *E. coli* may be due to differences in phytochemical composition and lower interaction efficiency with the bacterial outer membrane. Overall, both materials exhibit significant antibacterial properties, with AZ@ZIF-8 showing broader and more potent activity, highlighting the effectiveness of plant-loaded ZIF-8 composites as antimicrobial agents.

CHAPTER -5

SUMMARY

This work focuses on metal–organic frameworks (MOFs), especially ZIF-8, and their use in biomedical applications. ZIF-8 is chosen because of its high surface area, stability, and ability to carry drugs. The study also looks at how medicinal plant extracts, *Azadirachta indica* and *Acalypha indica*, can be loaded into ZIF-8 to combine the benefits of nanomaterials with natural bioactive compounds. *Azadirachta indica* extract is known for antibacterial, wound-healing, and anticancer properties, while *Acalypha indica* provides antioxidant and anti-inflammatory effects. By loading these extracts into ZIF-8, the framework not only keeps its porous structure but also gains added therapeutic functions. SEM analysis shows clear changes in particle morphology after loading, which confirms successful incorporation of the plant extracts into the ZIF-8 framework.

CONCLUSION:

The study proves that ZIF-8 is a strong and versatile carrier for medicinal plant extracts. Pure ZIF-8 shows uniform crystalline particles, while the extract-loaded samples show bigger, rougher, and more distinct particles. These changes mean that the phytochemicals are well integrated into the framework. Both *Azadirachta indica* and *Acalypha indica* extracts improve the biomedical potential of ZIF-8, but they influence the structure in different ways because of their unique phytochemical compositions. This difference is important, as it can affect drug loading, release behavior, and overall therapeutic performance. In simple terms, combining ZIF-8 with plant extracts creates a hybrid material that brings together modern nanotechnology and traditional medicine. This makes it a promising system for future drug delivery applications, with better efficiency and added natural bioactivity.

FUTURE SCOPE:

The present study shows that ZIF-8 can successfully carry medicinal plant extracts like *Azadirachta indica* and *Acalypha indica*. Still, there are many directions where this work can be extended.

- **Extended Plant Screening:** More medicinal plants with different phytochemical profiles can be tested with ZIF-8 to compare how their bioactive compounds affect framework morphology and drug release.
- **Detailed Release Studies:** Future work should focus on drug release kinetics under different pH and temperature conditions to confirm controlled release behavior in real biological environments.
- **In-vitro and In-vivo Studies:** Biological testing on cell lines and animal models will be important to validate the therapeutic efficiency and safety of these hybrid nanocomposites.
- **Combination Therapy:** ZIF-8 can be explored for co-loading of plant extracts with conventional drugs to achieve synergistic effects against resistant pathogens or cancer cells.
- **Advanced Characterization:** Techniques like TEM, BET surface area analysis, and FTIR can be used to further confirm encapsulation, pore changes, and chemical interactions between ZIF-8 and plant extracts.
- **Scalability and Applications:** Large-scale synthesis and formulation into capsules, gels, or coatings can be studied to move these materials closer to practical biomedical applications.

CHAPTER - 6

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